The Effect of Partisan Primaries on Turnout and Representation*

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Abstract

What are the representative and participatory consequences of primary participation rules? Elections rules that open participation to all eligible voters and allow more flexibility at the ballot box should boost turnout and reduce disparities in the primary electorate. However, some scholars argue that disparities in the primary electorate are not that large and that primary rules are ineffective in achieving these goals. Utilizing original panel data on state primary rules and a decade of nationwide voter file data comprising 2.6 billion observations, I use a difference-in-differences design to test the effects of opening primaries to unaffiliated voters on who participates. I find that primary electorates differ on important dimensions from general electorates and even more so from the pool of eligible voters. States increase voter turnout by an average of 5 percentage points when they open their primaries to unaffiliated voters and by nearly 11 percentage points when they switch to nonpartisan primaries. Additionally, the makeup of the primary electorate becomes more representative: younger, less party-affiliated, and more racially diverse. At a time when an unprecedented number of states are considering reforms to their primary systems, this research sheds light on the role that primary rules play in shaping who has democratic voice in America.

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1 Introduction

Do primary election reforms that allow unaffiliated voters to participate increase turnout and make the electorate more representative? Primary elections were instituted as a progressive-era reform aimed at greatly expanding democratic engagement (Hirano and Snyder 2019). However, these nominating contests have consistently attracted a fraction of the participation that general elections receive. Turnout averages only 20% of all eligible voters in recent primary election cycles (Ferrer and Thorning 2023). This lack of interest from voters raises concerns that the voters who do participate in primaries are unrepresentative of voters who participate in general elections or who are eligible to vote (Centeno et al. 2021). Primary elections have been blamed for contributing to many of our democracy's maladies, including polarization (Hall 2015), dysfunctional and unrepresentative government, declining approval of Congress, and negative campaigning (Troiano 2024).

In recent years, states have experimented with the rules governing primary participation, including opening primary participation to unaffiliated voters and instituting nonpartisan primaries. In the 2024 election cycle, voters weighed in on open and nonpartisan primaries in over half a dozen states.[1]. New Mexico opened their primaries to unaffiliated voters last year.[2] Now, a growing countertrend in conservative states has led Wyoming to close their primary elections to participation from voters not affiliated with a specific party, and Texas and Ohio are considering a similar move.[3]

Yet scholars disagree about the necessity (McDonald and Merivaki 2015; Sides et al. 2020) and the effectiveness (Centeno et al. 2021; Norrander and Wendland 2016) of these reforms.

This project uses original and large-scale administrative data to shed new light on how representative primary electorates are and to test the effectiveness of these reforms in increasing voter turnout and partisan, demographic, and racial diversity of the electorate. I

[1] https://www.nytimes.com/2024/06/25/us/ohio-redistricting-ballot-measures.html

[2] https://apnews.com/article/new-mexico-open-primaries-87d4d04bf0de858f2287f1d36b360b4e

[3] https://www.governing.com/politics/why-states-should-think-again-before-closing-primaries

create an original panel of state primary types going back to 2000 that categorizes differences between the two major parties and between presidential, congressional, state executive, and state legislative offices. I combine this with L2 nationwide voter file data for even-year elections between 2014 and 2024, a resource that contains billions of individual observations and can be used to accurately measure changes in turnout and demographic composition of the electorate over time. I also employ cutting-edge techniques for accurately estimating population-level racial demographics from individuals' name, geo-location, and turnout record (Imai, Olivella, and Rosenman 2022; McCartan et al. 2025).

I find in descriptive analysis that states have shifted from hosting closed primaries to primary contests that allow unaffiliated voters to participate and to nonpartisan primaries over the past two decades. Primary electorates include fewer younger, nonwhite, and unaffiliated voters than general election contests. Importantly, in every way that primary electorates differ from general electorates, they differ in a way that makes them *less* representative of the eligible electorate. I find stark differences between primary and eligible voters across nearly every dimension examined: racial diversity, age, party affiliation, income, education, and veteran status.

I leverage primary reforms in Alaska, Colorado, Idaho, Maine, and Oklahoma, Washington and a difference-in-differences design to test the causal effects of opening primaries to unaffiliated voters, opening primaries across party lines, and switching to nonpartisan primary systems on voter turnout and the composition of the electorate. When states reduce the barriers to primary participation, voter turnout sharply increases and the electorate grows more representative of the general electorate: younger, less party-affiliated, and possibly more racially diverse. The results are robust to a wide range of alternative specifications, including event study designs, heterogeneous-robust estimators, and randomization inference. This is the strongest evidence to date that primary rules shape who votes in meaningful ways.

2 Existing Literature

Representation in primaries matters because who shows up at the polls shapes who gets nominated and, ultimately, who gets elected. If institutional rules prevent all otherwise eligible voters from participating on Election Day, then the primary electorate might not match either the general election electorate or the broader pool of eligible voters.

Scholars have long worried that primary voters are unrepresentative of the general electorate (Key 1956; Ranney 1972)—in particular, that primary voters are more ideologically extreme and of higher socioeconomic status (Polsby 1983; Ranney 1975). This, in turn, may increase polarization by inducing congressional candidates to position themselves as ideologically extreme in order to win nomination (Brady, Han, and Pope 2007; Hall 2015; Hirano and Snyder 2019; King, Orlando, and Sparks 2016; Meisels 2026). Primary voter turnout is substantially lower than general election voter turnout and has been on the decline over several decades (Hirano 2010), although recent cycles have witness an upswing. Participation is particularly low in midterm, off-cycle, runoff, and special primaries. Only 21.3% of eligible voters participated in the 2022 primary elections and only 14.3% participated in the 2014 primary cycle (Ferrer and Thorning 2023). 80% of eligible voters regularly do not participate in selecting which nominees appear in the general election.

However, scholarship is split on the extent of demographic and ideological disparities between primary and general electorates. Some argue that primary voters tend to have similar demographic attributes and policy attitudes to the party faithful (Geer 1988) or even rank-and-file voters Sides et al. (2020). On the other hand, others argue that primary voters are indeed more ideologically extreme and demographically distinct from general election voters Hill and Tausanovitch (2018); McDonald and Merivaki (2015).

Another contested question is the effect of different primary rules on voter participation and the demographic and ideological composition of the electorate. Some studies find little relationship between primary type and ideological orientation of the electorate (McDonald and Merivaki 2015; Norrander and Wendland 2016; Sides et al. 2020), voter turnout (Bonneau and Zaleski 2021), or legislature extremism (Crosson 2021; McGhee and Shor 2017). However, other research concludes that primary type does affect voter turnout (Donovan, Micatka, and Tolbert 2025; Geras and Crespin 2018; Jewitt 2019; Micatka et al. 2024), the partisan composition of the electorate, (Micatka et al. 2024), voter's party identification (Burden and Greene 2000; Finkel and Scarrow 1985; Norrander 1989), and the ideological extremism of legislators (Anderson et al. 2023; Gerber and Morton 1998; Wright, Reilly, and Lublin 2025). Primary type can also shape the demographics of the electorate in other ways. Open and nonpartisan primaries are associated with younger voters (Donovan, Micatka, and Tolbert 2025; Kaufmann, Gimpel, and Hoffman 2003), whereas closed primaries are associated with reduced levels of participation by people of color, especially Asian Americans and Latino voters (Centeno et al. 2021).

3 Data and Methods

I create original state-level panel of primary type rules between 2000 and 2024 for each regularly scheduled primary.[4] The data for this categorization comes from a mix of sources: previous work by Unite America and the National Conference of State Legislatures (NCSL), contemporaneous newspaper reports, archived government websites, and direct communication with state and party officials. Rather than identifying one primary rule for all of the elections in each state, I distinguish Democratic and Republican party primary rules for presidential, congressional, state executive, and state legislative offices. I count a primary election for state executive office as occurring if it includes at least one of the following offices: governor, lieutenant governor, secretary of state, attorney general, state treasurer, state comptroller/controller, state auditor, superintendent of public instruction, agricultural commissions, insurance commissioner, or elected commissioners for labor, mine inspector, land, or tax.[5] For states that elect some executive offices in one year and some executive

[4] Louisiana's jungle primaries are excluded because they can take the place of a general election.
[5] I do not include executive primaries for state board of education or any other state, local, or tribal offices.

offices in a different year, I do not distinguish the specific offices elected but count both years as constituting a state executive primary. I do not distinguish whether the primary was contested, but do exclude cases where the primary was canceled altogether (as has happened with some recent presidential primaries do to COVID-19), as well as cases where a state party convention made nominating decisions even if a nonbinding or "beauty pageant" primary was held. When states hold both a caucus and a primary for president, I count the rules for which the primary was held. When a state only holds a caucus, I identify the rules used for the caucus. When states hold multiple primaries for the same category in the same year (i.e., New York holding separate primaries for Senate and House elections in 2022), I consider the rules used for the primary that came first in the calendar year.

I categorize states into five types based on their rules for primary voter participation. States with "closed" primaries only allow voters with party affiliation to vote in that party's primary. They do not allow voters with prior party affiliation to vote in a different party's primary, nor do they allow voters not registered with a party to vote for partisan offices. States that are "Open To Unaffiliated" allow voters not registered with a party to vote for the party of their choice, but do not allow voters registered with a party to vote in a different party's primary. States that are "Partially Open" allow both unaffiliated voters and voters registered with a party to vote for the party of their choice. However, that choice registers the voter with that party on the voter file. "Open" primaries are similar to partially open ones, with the difference being that selecting a party's ballot does not register the voter with that party. Finally "nonpartisan primaries" (also called "multi-party primaries") group candidates from all parties together on the same ballot. All voters can therefore vote for the candidate of their choice for each office and may select candidates from different parties for different offices. Different variations of nonpartisan primaries include blanket, top-two, and top-four primary systems. In blanket primaries, the top vote-getter from each party advances to the general election. In top-two and top-four systems, the top X number of candidates with the most votes advance, regardless of party affiliation. Therefore, the general election might feature multiple candidates with the same party affiliation. Alaska's top-four system includes an additional innovation–ranked choice voting in the general election. This helps reduce the incidence of strategic voting and disproportionate outcomes, whereby the majority party splits its votes between two candidates, allowing the less numerous party to get their candidate elected.

I use L2 nationwide voter file data for the descriptive and statistical analysis of turnout and electorate composition. L2 is a private company that combines each state's voter file and adds a range of demographic and commercial data. I have access to this data for primary and general elections that took place between 2014 and 2024. The L2 voter file contains millions of observations for each election and billions of observations in total. It is the best data source available for information on who voted in each election, as it is not susceptible to sampling or non-response bias. Following best practices for L2 voter file use (Kim and Fraga 2022), I list the file date used to capture each state-election in Section A.2 in the online appendix.

From the voter file, I derive the number of voters and registrants for each election. I impute race in two stages. First, I implement a fully Bayesian Improved Surname Geocoding (fBISG) using census block geocodes where possible (otherwise, tract, zip code, county, or surname-only, in that order), as well as registered voters' full name and party affiliation (Imai and Khanna 2016; Imai, Olivella, and Rosenman 2022). The fBISG method improves on BISG by accounting for census measurement error and surnames that are missing from the census data. Second, I use Bayesian Instrumental Regression for Disparity Estimation (BIRDiE) to accurately estimate turnout by racial groups (McCartan et al. 2025). I group turnout data by election date and run a Categorical Dirichlet model, with state-level random effects included for dates with contests in across multiple states. I estimate the posterior probabilities that each individual registrant is non-Hispanic white, Black, Asian, Hispanic, or Other.

I use citizen voter-age population (CVAP) estimates from the ACS 5-year reports as the denominator in calculations of voter turnout.[6] I calculate racial composition shares as the number of voters of a certain race (summed probabilities) divided by the total number of voters with racial data available. Nonwhite share is calculated as the share of voters that are not non-Hispanic white. I use L2 registration and commercial records to calculate the median age of voters at the time of each election; the share of voters that are Democratic, Republican, Third-party, and unaffiliated;[7] the share of voters that are female; the share of voters that make less than $50,000 a year, less than $100,000 a year, and more than $250,000 a year; the share of voters with at least some college education; the share of voters who are working-class; and the share of voters that are veterans.[8]

I use L2 data to calculate turnout and demographics for both primary and general elections, as well as the differences between the two. I also use L2 to calculate the electorate composition of registered voters and data from the 2018 Cooperative Election Survey (CES) to calculate the electorate composition of all citizen voting-age eligible individuals.[9]

I estimate the causal effects of opening primaries on electorate demographics by leveraging states that have switched their primary type over the past decade. I employ a difference-in-difference estimation with state and year fixed effects. This design accounts for the fact that states that use open primaries might have different electorates than those that use closed primaries for reasons beyond the specific primary type employed, and thus a simple comparison between open and closed primary states produces a biased estimate of the causal effects of primary type on representation. My preferred specification includes state by party by office and year fixed effects, which compares turnout and electorate composition between

[6] `https://www.census.gov/programs-surveys/decennial-census/about/voting-rights/cvap.html`

[7] L2 party affiliation data varies by state. L2 uses official voter registration in the majority of states that record party in the voter file. In states that do not record party affiliation, L2 uses primary ballot choice where possible and otherwise relies on modeling analytics that takes into account demographic, commercial, political contribution, and other election data.

[8] Working-class voters are defined per Carnes (2016) as those who work in food services, laborer, maintenance, manufacturing, office assistant, sales, skilled trades, or transportation.

[9] I subset to respondents who are American citizens and employ the post-stratification weights provided in the survey data. This source allows me to calculate estimates of the pool of eligible voters across all racial, political, and demographic variables, except for the share of eligible voters that are veterans.

elections at the state-party-office level. For instance, this analysis compares the primary electorate in Idaho's Republican presidential primary in 2020 to the electorate in Idaho's Republican presidential primary in 2024, when participation was opened up to unaffiliated voters. For primaries with more than one contest category on the ballot (e.g., president, Congress, state executive, and state legislative) or Democratic and Republican primaries held at the same time, multiple state-party-office observations share the same underlying turnout and demographics. Therefore, the design gives equal weight to the rules for each party-contest. This allows me to take into account variation in primary rules by both contest and party within each state.

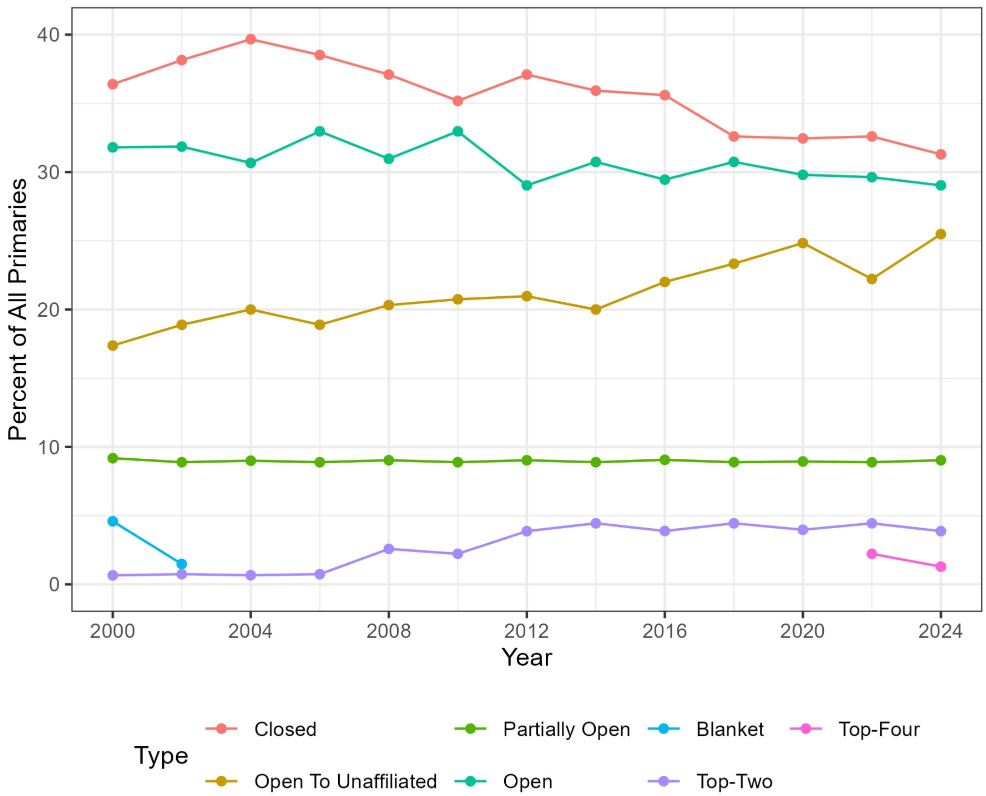
My approach builds on existing literature in several ways. First, almost all previous research has been survey based. These studies utilize a relatively small number of observations, typically between 1,000 and 10,000. Even the largest studies sample fewer than 100,000 people. In contrast, the voter file includes administrative records on every registered voter and amounts to hundreds of millions of observations for each election. Surveys are imperfect because samples can produce noisy estimates, especially for relatively small demographic groups, and because surveys may produce bias. Some surveys eliminate social desirability bias by validating turnout using the voter file, but sampling and non-response bias remain of concern (Grimmer et al. 2018). Administrative data solves these problems, since the sample is the complete population of registered voters. Second, almost all previous research only examines one point in time or spans a handful of elections. The only existing scholarship that utilizes administrative data relies on one cross-sectional slice of the voter file Donovan, Micatka, and Tolbert (2025); Micatka et al. (2024). My data spans every primary and general election that took place between 2014 and 2024. Third, the panel nature of the voter file allows me to credibly estimate the effects of primary type on the composition of the electorate, relying on weaker inferential assumptions than have been made in previous work.

3.1 Changes in State Primary Type, 2000-2024

Figure 1 shows the percentage of primary elections with each primary type in even-year elections between 2000 and 2024. The data is at the state-office-party level, which means the figure captures differences in rules between the Democratic and Republican parties and across presidential, congressional, state executive, and state legislative offices. The percentage of primary elections that are fully closed to those not registered with a specific party has declined over time. In 2000, 36% of all primary elections were closed to unaffiliated voters. By 2024, this has dropped to 31% of all elections. The percentage of primaries that are open to registered voters irrespective of party affiliation has also slightly diminished, from 32% in 2000 to 29% in 2024. Opening up primaries to unaffiliated voters, on the other hand, has become much more popular over the past two decades, rising from 17% to 25% of all elections. Finally, more states are experimenting with nonpartisan primaries. In 2000, three states used some form of nonpartisan primaries: California, Nebraska and Washington. Alaska joined this group in the 2022 primary cycle, and several states had nonpartisan primary initiatives on the ballot in 2024.

Section A.1 in the online appendix visualizes changes in primary type by party and region. Republicans are more likely to use closed primaries, whereas Democrats are more likely to open their congressional primaries to unaffiliated voters. Both major parties have seen a shift from closed to open to unaffiliated primary rules over the past quarter century, though the trend is more dramatic for Democratic primaries. Most primary reforms have occurred in New England and Western states.

Figure 1: **Type of Presidential, Congressional, State Executive, and State Legislative Primaries, 2000-2024**. This graph displays the percentage of primary elections with each primary type in even-year elections between 2000 and 2024. The data is at the state-office-party level, which means the figure captures differences in rules between the Democratic and Republican parties and across presidential, congressional, state executive, state legislative offices.



4 Descriptive Results

This section examines descriptive evidence for the representativeness of primary electorates and differences in turnout and electorate demographics by primary type.

4.1 Do Primary Electorates Resemble General Election and Eligible Voter?

Is the primary electorate unrepresentative of general election or eligible voters? Table 1 shows the average turnout and compositional shares of the primary and general election electorates, as well as the difference between the two. I also show the composition shares of the pool of registered voters and the pool of eligible voters.

Table 1: Turnout and Composition of Primary and General Electorates, 2014–2024

Electorate	Primary	General	Difference	Registered	Eligible
Turnout					
Turnout (overall)	21%	53%	-32%	-	-
Black turnout	11%	33%	-22%	-	-
Latino turnout	6%	26%	-19%	-	-
Asian turnout	8%	34%	-26%	-	-
White turnout	24%	59%	-35%	-	-
Racial Demographics					
Share Nonwhite	15%	18%	-3%	23%	26%
Share Black	8%	8%	0%	10%	11%
Share Latino	3%	5%	-1%	6%	8%
Share Asian	2%	2%	-1%	3%	3%
Share Other	2%	2%	-1%	4%	4%
Share White	85%	82%	3%	77%	74%
Other Demographics					
Median age (years)	61	55	6	51	48
Share Democratic	43%	37%	6%	36%	32%
Share Republican	47%	39%	8%	34%	29%
Share 3rd Party	1%	2%	-1%	2%	4%
Share Unaffiliated	10%	22%	-13%	28%	34%
Share Female	54%	53%	1%	53%	51%
Share under 50k income	28%	26%	1%	28%	45%
Share under 100k income	76%	76%	1%	77%	73%
Share over 250k income	2%	2%	0%	2%	1%
Share some college	68%	68%	0%	67%	62%
Share working-class	28%	30%	-1%	30%	-
Share veteran	5%	5%	1%	4%	12%

Primary, General, and Registered data come from L2 data. Primary, General, and Registered race data is estimated using Imai et al. (2022)'s fBISG method at the block-group level with full name and party data. These are then updated with the BIRDiE method (McCarten et al. 2025), using observed voting outcomes and state-level random effects for multi-state elections. "Eligible" is defined as CVAP, or the "citizen voting-age population", which counts as eligible voters all those who are adults and U.S. citizens. It is sourced from the 2018 Congressional Election Survey. Difference is Primary − General. All data is averaged at the state-election date level. The share of eligible voters that are working-class is not available in the CES data.

In line with previous scholarship, voter turnout rates are much lower in primary elections than in general elections (Ferrer and Thorning 2023). On average, about one in five eligible voters participate in the average state primary election, compared with over half of eligible voters participating in a typical state's general election. Primary turnout rates are lower than general election turnout rates across racial and ethnic groups. However, proportionally fewer racial minorities vote in primaries compared with general elections. Sixty percent of CVAP white voters participate in general elections, compared with 25% in primaries. This means white primary voter participation is 42% of white general election participation. Only 11% of eligible Black voters vote in primary elections compared with 33% in general elections, meaning Black primary election turnout is 37% that of Black general election turnout. Latino primary participation is 27% of general election participation, and Asian primary participation is only 23% of Asian general election participation. In the average state, only 11% of eligible Black voters participate in primaries, and fewer than 1 in 10 eligible Latino and Asian voters participate. These differential turnout rates contribute to primary electorates that are not racially reflective of the general, registered, or eligible electorates.

In terms of composition of the electorate, primary and general election voters are similar across a number of categories, including gender, income, education, and veteran balance. Working-class voters are slightly underrepresented in primaries compared to general elections. Primary electorates differ substantially with general electorates on two dimensions: age and party affiliation. The primary electorate is much older and more likely to be party affiliated. The median age of voters in primary elections is 61 compared with 55 in general elections. This is unsurprising considering that older voters are more likely to be habitual voters (Plutzer 2002). Primary electorates are also much more major-party affiliated than general electorates. The average primary electorate in a state between 2014 and 2024 was 43% Democratic party-affiliated, 47% Republican party-affiliated, 1% Third-party affiliated, and 10% unaffiliated. This contrasts with 22% of general election voters who are unaffiliated, on average.

Columns 4 and 5 compare primary and general election voters to the universe of registered and eligible voters. Across virtually every category, the primary electorate is less representative of the pool of citizen voting-age eligible voters than the general electorate, with the pool of registered voters in-between the general and eligible electorates. Nonwhites make up 25% of the pool of eligible voters and 23% of all registered voters in the average state, but only 17% of the general election electorate and 15% of the primary electorate. The median age of eligible voters is 49, compared with 51 among registered, 55 among general election voters, and 61 among primary election voters. Unaffiliated voters comprise 34% of the average state's pool of eligible voters and 28% of its pool of registered voters, but only 22% of the general electorate and 10% of the primary electorate. Low-income, working class, veterans, and non-college-educated voters are all underrepresented in the primary and general electorates compared the pool of registered and eligible voters. In so far as an electorate representative of the pool of eligible voters is a normative ideal—which many foundational works in the discipline posit (Dahl 1989; Lijphart 1997; Schattschneider 1960; Verba and Nie 1972; Verba, Schlozman, and Brady 1995)—the primary electorate is even further off the mark than the general electorate.

Table A.2 in the appendix shows that differences in the average state's turnout and composition of the primary versus general electorate are consistent across years. Overall primary turnout is far lower than overall general election turnout, ranging from a 23 percentage point difference in 2014 to a 45 percentage point gap in 2024. Turnout among different racial and ethnic groups are also consistently lower in primary than general elections. Turnout gaps are largest in presidential election cycles, especially since many states choose to hold their presidential and congressional/state primaries on different days and general presidential election turnout far exceeds general midterm turnout. In terms of the makeup of the electorate, racial minorities, unaffiliated, younger, and working class voters are also consistently underrepresented in primary elections.

In summary, primary electorates significantly differ from general electorates and the pool of eligible voters, especially in terms of party affiliation, race, and age, and class. Primary electorates are whiter and older than general election and eligible voters. They are also substantially more likely to affiliate with a major political party. Across all racial, economic, and demographic characteristics studied, primary electorates are worse approximations of the pool of eligible voters than general electorates. This means it is worth studying the effects of reforms designed to increase turnout and make the primary electorate more closely resemble the pool of eligible voters.

4.2 Differences in Turnout and Electorate Composition by Primary Type

How does turnout and demographic composition of primary electorates vary by primary type, relative to general electorates? In other words, under which rules do primary election voters look most similar to general election voters? Table 2 compares primary and general electorates, grouping states by the rules they use to determine who can participate in their congressional primaries. Columns with hyphenated primary types indicate that the Democratic and Republican parties use different rules for congressional contests. All numbers show the gap in turnout or compositional share between primary and general elections, according to the rules used by each state which dictate who can participate in the state's congressional primaries that year. Turnout gaps between primary and general elections tend to be smaller under open or nonpartisan primary rules than under closed primary rules. The average gap in turnout between primaries and general elections under closed primary rules is 32 percentage points, compared with 28 percentage points under open and top-four primaries and 22 percentage points under top-two primaries. The same is true of turnout rates for most racial and ethnic groups. The gap in primary turnout for Latinos is 20 percentage points on average under closed primary rules but is 14–18 percentage points under open or nonpartisan primary rules. The gap in Asian turnout between primary and general elections is typically 26 percentage points under closed rules, but is only 13 percentage points under top-four primaries.

Table 2: Differences in Turnout and Composition Between Primary and General Electorates By Primary Type, 2014–2024

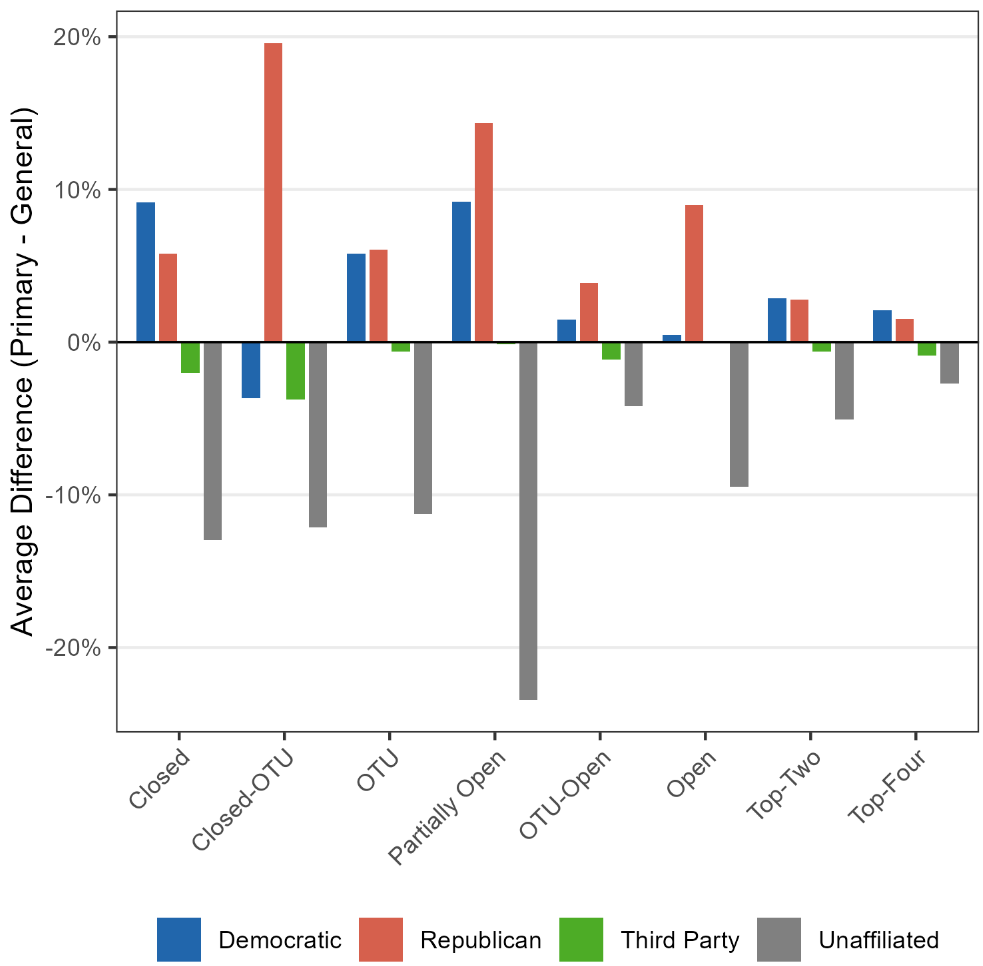
	Closed	Closed-OTU	OTU	Partially Open	OTU-Open	Open	Top-Two	Top-Four
Turnout								
Turnout	-32%	-28%	-30%	-35%	-32%	-28%	-22%	-28%
Black turnout	-21%	-16%	-18%	-22%	-22%	-21%	-16%	-15%
Latino turnout	-20%	-11%	-18%	-20%	-18%	-14%	-18%	-15%
Asian turnout	-26%	-15%	-26%	-31%	-19%	-21%	-20%	-13%
White turnout	-36%	-30%	-33%	-38%	-38%	-31%	-22%	-34%
Racial Demographics								
Share Nonwhite	-2%	-2%	-2%	-2%	-3%	-3%	-7%	-5%
Share Black	1%	0%	0%	0%	-1%	-1%	-1%	-1%
Share Latino	-1%	-1%	-1%	-1%	-1%	-1%	-4%	-1%
Share Asian	-1%	0%	-1%	-1%	-1%	0%	-1%	-1%
Share Other	-1%	-1%	0%	0%	-1%	-1%	-1%	-2%
Share White	2%	2%	2%	2%	3%	3%	7%	5%
Other Demographics								
Median age of voters	5	5	5	6	4	5	5	3
Share Democratic	9%	-4%	6%	9%	1%	0%	3%	2%
Share Republican	6%	20%	6%	14%	4%	9%	3%	2%
Share 3rd Party	-2%	-4%	-1%	0%	-1%	0%	-1%	-1%
Share Unaffiliated	-13%	-12%	-11%	-23%	-4%	-9%	-5%	-3%
Share Female	1%	1%	0%	0%	1%	0%	0%	1%
Share under 50k income	3%	2%	2%	1%	0%	1%	1%	-
Share under 100k income	1%	1%	2%	0%	-2%	0%	1%	-
Share over 250k income	0%	0%	0%	0%	0%	0%	0%	-
Share some college	0%	0%	0%	0%	0%	0%	0%	1%
Share working-class	-1%	-2%	-1%	-1%	-2%	-1%	-1%	-1%
Share veteran	1%	1%	1%	1%	1%	1%	1%	1%

Each cell shows the average state primary turnout/electorate composition for that primary type minus the average state general election turnout/electorate composition. For share rows, positive values indicate over-representation in the primary election relative to the general election and negative values indicate under-representation. All data is averaged at the state level.

In terms of composition of the electorate, there is a clear trend in partisan affiliation, illustrated in Figure 2. Under closed primaries, registered Democrats and Republicans are over-represented by 9 percentage points and 6 percentage points, respectively, compared with the electorate in the general election. In contrast, third-party voters and unaffiliated voters are underrepresented by 2 and 13 percentage points, respectively. These imbalances tend to diminish as primary participation rules become more lenient with regard to party affiliation.

These comparisons are suggestive rather than causal in nature, since states that happen to have open or nonpartisan primary turnout could attract significantly different primary participation and electorate compositions for reasons besides the participation rules themselves. In the next section, I turn towards a more inferentially sound design testing the effects of primary reform on voter turnout and electorate composition.

Figure 2: **Average Difference In Partisan Composition Between Primary and General Electorates by Congressional Primary Type**. Each bar is the average gap in partisan composition between primary and general electorates among states that use the specified primary type. Positive numbers indicate voters registered with that party affiliation are over-represented in primary elections relative to general elections. Negative numbers indicate voters of that affiliation are underrepresented in primaries.



5 Does Opening Primaries Increase Voter Turnout and Improve Representation?

In this section, I leverage primary reforms to the rules for participation between 2014 and 2024 to run a difference-in-differences design, comparing changes in turnout and electorate composition of a state that has undergone reform to changes in states that have not reformed their primary systems. Primary reforms implemented between 2014 and 2024 were most commonly switches between closed primary and open-to-unaffiliated primary systems, but also include switches between closed and open, open, between open to unaffiliated and top-four, and between open and top-four. Therefore, regressions capture the effect of switching from a closed primary rule to an open-to-unaffiliated, open, or top-four primary rule on turnout and composition of the electorate.

Analyses are powered by reforms in six states: Alaska, where all but Presidential contests switched to a top-four system in 2022 (from an open rule for Democrats and an open-to-unaffiliated rule for Republicans); Colorado, which opened their primaries to unaffiliated voters in 2018; Idaho, whose Republican party allowed unaffiliated voters to voter for state executive and legislative elections in 2018 and whose Democratic party stopped allowing unaffiliated participation; Maine, which switched their congressional and state legislative primaries from closed to open-to-unaffiliated in 2024; Oklahoma, whose Democratic party opened up their congressional and state legislative primaries to unaffiliated voters in 2016 and opened their state executive elections to unaffiliated voters in 2018; and Washington, where Republicans opened their presidential primary from just Republicans to all voters in 2020.

I begin by studying the effect of primary reform on voter turnout and on the partisan, racial, and demographic composition of the electorate. I then validate the robustness of these results to different samples, units of analysis, heterogeneous-robust estimators, parallel trends scrutiny, and placebo tests. Finally, I examine the effect of reform on participation disparities between same-year primary and general elections and on party registration of registered voters. All analyses are conducted at the state-party-office-election level unless otherwise noted.

5.1 Impact of Open Primaries on Voter Turnout

Primary election voter turnout is very low. Fewer than 20% of eligible voters have participated in recent midterm primary elections, far below participation rates of general elections(Ferrer and Thorning 2023). When states or parties open their primaries to unaffiliated or cross-party voters, does this boost participation? Table 3 shows the results of difference-in-differences regressions on the effect of switching from a closed primary rule to an open-to-unaffiliated, open, and top-four primary system on voter turnout. Regressions are run at the state-party-office-year level, capturing the fact that primary types vary within states by party and office. The reference category is turnout in closed primary elections. Column 1 shows that overall primary voter turnout increases by 4.9 percentage points, on average, when states allow those without formal party affiliation to participate in the nominating process. Columns 2 through 5 examine changes to turnout levels of specific racial and ethnic groups. Participation increases across racial groups, with Black voters and Latino voters seeing a 2 percentage point increase, Asian voters a 3 percentage point increase, and white voters a nearly 6 percentage point boost in participation.

The effects are even larger when switching to an open or nonpartisan primary system. Opening primary participation to cross-party voters increases voter turnout by 6 percentage points on average. Switching from closed elections to a top-four nonpartisan system boosts overall voter turnout by nearly 11 percentage points. In each case, all major racial/ethnic groups see a sizable boost in their turnout rates, though white voters tend to see the largest absolute increases in voting.

Table 3: **Opening Primaries Increases Voter Turnout**

	Voter Turnout				
	Overall (1)	Black (2)	Asian (3)	Latino (4)	White (5)
Open To Unaffiliated	0.049	0.021	0.030	0.022	0.058
	(0.015)	(0.006)	(0.009)	(0.006)	(0.017)
Open	0.061	0.028	0.040	0.025	0.072
	(0.015)	(0.009)	(0.011)	(0.006)	(0.017)
Top-Four	0.107	0.021	0.043	0.045	0.148
	(0.015)	(0.008)	(0.010)	(0.006)	(0.017)
Observations	1680	1680	1680	1680	1680
States	50	50	50	50	50
Years	8	8	8	8	8
Outcome Mean	0.213	0.105	0.080	0.063	0.246
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Voter turnout is measured as a proportion out of 1 using citizen voting age population estimates from the U.S. Census. The omitted primary type is closed. The data range is 2014–2024.

5.2 Impact of Open Primaries on the Primary Electorate

Does opening primaries to unaffiliated or cross-partisan voters or switching to a non-partisan primary system lead to more ideologically and demographically representative electorates? Table 2 showed that states with more open primary types tend to have primary electorates that are more representative of the general election electorate. However, this relationship could be correlational rather than causal in nature. For example, states that have more open primary types could also happen to have more diverse demographics. Here I examine what happens to the composition of a state's primary electorate when it starts allowing unaffiliated or cross-partisan voters to participate in these contests.

Table 4 shows the effect of opening primaries on the partisan composition of the electorate. The unaffiliated share of the electorate increases by 14 percentage points. This increase comes almost entirely at the expense of the share of voters who are affiliated with the Democratic and Republican parties–whose share decreases by 5 and 9 percentage points, respectively. In other words, as the share of unaffiliated and third-party voters increases, Democratic and Republican shares decrease. Similar effects are seen for a switch to open (10 percentage point boost in unaffiliated share) and nonpartisan (13 percentage point boost in unaffiliated share) primary systems, with major reductions in the share of Democratic- and Republican-affiliated voters in most cases.

Table 4: **Effect of Opening Primaries on Partisan Composition of Electorate**

	Partisan Share			
	Unaffiliated (1)	Third-party (2)	Democratic (3)	Republican (4)
Open To Unaffiliated	0.138	-0.002	-0.048	-0.088
	(0.030)	(0.001)	(0.012)	(0.028)
Open	0.102	-0.002	-0.018	-0.083
	(0.036)	(0.001)	(0.026)	(0.021)
Top-Four	0.133	0.003	0.005	-0.142
	(0.031)	(0.001)	(0.018)	(0.025)
Observations	1680	1680	1680	1680
States	50	50	50	50
Years	8	8	8	8
Outcome Mean	0.102	0.008	0.416	0.474
State $\times$ Party $\times$ Office FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table 5 examines the effect of changing primary rules on the racial and ethnic composition of the electorate. In general, opening primaries has at most modest effects on the racial composition of the electorate, though Latino participation rates are consistently boosted when switching from closed voting rules.

Table 6 examines the effects of primary reform on a range of other demographic characteristics. When states switch to more open primary rules, the share of male voters increases by about 1 percentage point and the median age of voters decreases by between 1 and 2 years. It also seems to boost the share of the electorate that is low-income, and may modestly increases the share of the electorate without any college education. It does not appreciably alter the share of voters who are working-class or veterans.

Table 5: **Effect of Opening Primaries on Racial Composition of Electorate**

	Racial Share				
	Black (1)	Asian (2)	Latino (3)	Nonwhite (4)	White (5)
Open To Unaffiliated	-0.001	0.000	0.002	0.004	-0.004
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
Open	-0.000	0.002	0.003	0.009	-0.009
	(0.001)	(0.002)	(0.001)	(0.004)	(0.004)
Top-Four	0.004	-0.000	0.007	-0.014	0.014
	(0.001)	(0.001)	(0.001)	(0.003)	(0.003)
Observations	1680	1680	1680	1680	1680
States	50	50	50	50	50
Years	8	8	8	8	8
Outcome Mean	0.068	0.018	0.035	0.139	0.861
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table 6: **Effect of Opening Primaries on Demographic Composition of Electorate**

	Male Share (1)	Median Age (2)	Low-Income Share (3)	WC Share (4)	Low-Edu Share (5)	Veteran Share (6)
Open To Unaffiliated	0.012	-1.524	0.011	-0.001	0.003	-0.005
	(0.002)	(0.415)	(0.003)	(0.001)	(0.002)	(0.002)
Open	0.014	-1.938		-0.002	0.009	-0.004
	(0.002)	(0.511)		(0.002)	(0.005)	(0.004)
Top-Four	0.009	-1.200		-0.007	0.002	0.005
	(0.003)	(0.477)		(0.002)	(0.003)	(0.004)
Observations	1680	1680	553	1392	1392	1392
States	50	50	50	50	50	50
Years	8	8	3	7	7	7
Outcome Mean	0.462	61.683	0.282	0.288	0.682	0.056
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. The omitted primary type is closed. The data range is 2014–2024. Income data is only available for 2014–2016. Occupation, education, and veteran data is available from 2014–2022.

5.3 Validating the Effect of Open Primaries on Turnout and the Electorate

I validate four effects of opening primaries identified in the previous section: increased primary voter turnout, increased unaffiliated share of the primary electorate, increased nonwhite share of the primary electorate, and decreased median age of the primary electorate. I conduct five validation exercises: varying the sample, varying the unit of analysis, employing a heterogeneous-robust estimator, examining event-studies plots for parallel trends, and finally conducting randomization inference to verify a null placebo effect. The sum of evidence is that opening primaries to unaffiliated voters has a large positive effect on voter turnout and the unaffiliated share of the electorate and a large negative effect on the average age of the electorate. The evidence for an increase in racial diversity is mixed.

Section A.4 in the online appendix subsets the sample to state-party-offices that either always use closed elections ("true" controls) or switched from closed to open to unaffiliated. Sections A.5 and A.6 simplify the analysis to the state-election level by subsetting to only Congressional Democratic and Congressional Republican offices, respectively. Section A.7 addresses the potential biases in the difference-in-differences analysis due to the staggered nature of treatment (Callaway and Sant'Anna 2021; de Chaisemartin and D'Haultfœuille 2020; Goodman-Bacon 2021) by employing dynamic Callaway and Sant'Anna estimators. In all cases, the results are generally in line with those reported in the main analysis.

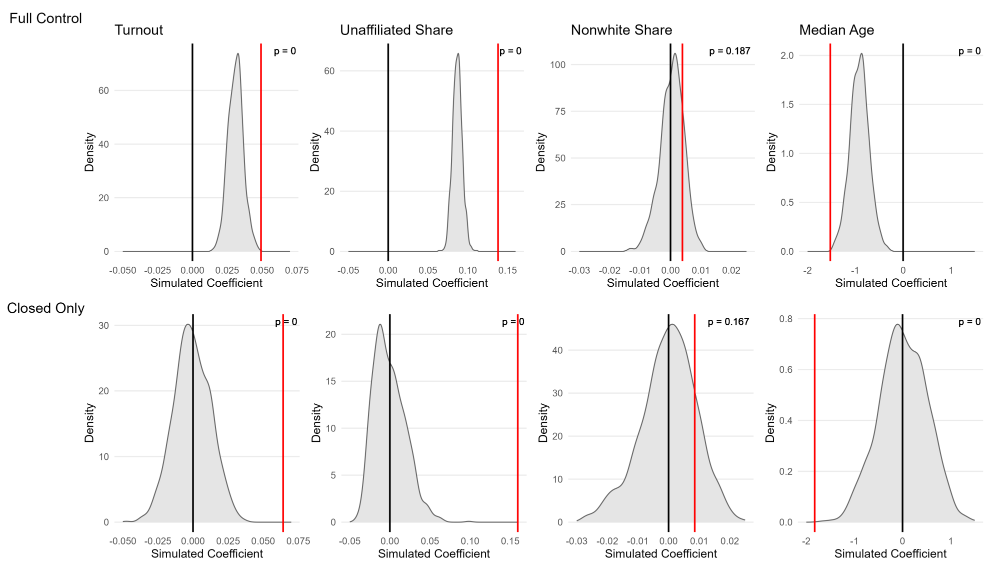
DID designs are dependent for causal inference on the parallel trends assumption—in order words, that states implementing primary reforms were on similar primary turnout and electorate composition trajectories as states that did not implement reforms. I conduct event studies estimators in Section A.8 that examine the effect of reform both prior to implementation and afterward. I generally find no evidence of problematic pre-trending for turnout, unaffiliated share, or age of the electorate, though I do find some evidence for pre-trending of nonwhite share of the electorate.

Finally, the results derive from a relatively small number of primary reforms. Are the observed effects simply due to the structure of the data itself rather than the primary reforms? I use randomization inference as a placebo test to estimate the likelihood of finding effects as large or larger than the ones observed by chance. In Figure 3, I employ randomization permutation on primary reforms switching for closed to open-to-unaffiliated using two different control samples: one including all other primary types (the top panel), and one subsetting to state-party-offices that are always closed. In both cases, I preserve the timing of primary reforms but randomly permute which treated units receive treatment. 1,000 permutations are computed for each exercise. In the top panel, the Open To Unaffiliated point estimates shown in Table 3 column 1, Table 4 column 1, Table 5 column 4, and Table 6 column 2 are replicated with the permuted data and the coefficient stored for each permutation. Finally, the actual coefficient derived is compared with the distribution of permuted coefficients. The p-value is the number of randomized coefficients that are greater than or equal to the actual estimated effect (in the expected direction) divided by the total number of iterations. The bottom panel point estimated can be found in Section A.4.

The results in the top panel shows that random treatment assignment of the treated units typically results in a positive relationship between open-to-unaffiliated rule and voter turnout, unaffiliated share, and median age, evidenced by the non-zero averages of the permutation. However, it is still extremely unlikely to get an observed effect as large as that actually observed for these outcomes—1 in 500 or fewer in all three cases. We cannot rule out the effect on increased nonwhite share to have arisen purely by chance.

The top panel generally produces biased permutations because this procedure assigns the treatment (open-to-unaffiliated primary rule) to state-party-offices that, in control, have open or nonpartisan rules rather than closed ones. It is reassuring that, even under the biased baseline (with the permutation averages deviating from zero), the estimated effects are still statistically significant for turnout, unaffiliated share, and median age.

Figure 3: **Randomization Inference for Effect of Primary Elections on Turnout and Electorate Composition**. This graph displays the output of randomization inference for the effect of switching from closed to open-to-unaffiliated primary rules on voter turnout, the unaffiliated share of the electorate, the nonwhite share of the electorate, and the median age of the electorate. Which state-office-parties receive treatment is randomly permuted, preserving the timing pattern of treatment. The black distributions shows the resulting coefficients of 1,000 iterations. The red solid vertical line is the actual coefficient observed, and the p-value is the share of coefficients that are equal to or larger than the one estimated in the respective specification. The top panel includes all control state-office-party units and corresponds to the Open To Unaffiliated point estimates in Table 3 column 1, Table 4 column 1, Table 5 column 4, and Table 6 column 2, respectively. The bottom panel includes state-office-parties that always use closed primaries as controls, with point estimates found in Section A.4.



The bottom panel corrects this bias by only using "clean" controls—those state-party-offices that in actuality always use a closed rule. This is evidenced by the density plots for all four outcomes centering around 0. The observed effect is more extreme than practically every simulated permutation in the case of turnout, unaffiliated share, and median age. This provides additional validation that opening primary elections positively affects turnout and partisan and demographic representativeness.

5.4 Impact of Open Primaries on Disparities Between Primary and General Electorates

Table 1 revealed significant differences in the composition of state primary electorates compared to general electorates. Do primary election reforms reduce disparities in turnout and demographic composition between primary and general election voters? In order to study this, I collapse the data to the state-year level. For states with multiple primaries in the same year, I take turnout and demographics from the first primary election in that year. I then calculate the disparity between regularly scheduled primary elections and the corresponding general election attributes for each state-year-office. Table 7 shows the effects on participation disparities and Tables 8, 9, and 10 shows the effects on composition disparities in party, race, and other demographics, respectively.[10],

The evidence presented in these tables suggest that opening primaries to unaffiliated voters significantly reduces participation and compositional disparities between primary and general elections. Table 7 shows that when states open their primaries up to unaffiliated voters, turnout disparities between primary and general election turnout decrease by 7.1 percentage points on average. Switching to an open primary rule reduces the primary-general turnout disparity by 7.9 percentage points, and switching to a top-four nonpartisan primary system reduces the turnout disparity by 17 percentage points. All major racial and ethnic groups see a significant decrease in their participation disparities under more open primary rules. Switching to more open primaries results in a 5 to 12 percentage point decrease in the Black turnout disparity, a 4 to 12 percentage point decrease in the Asian turnout disparity, and a 4 to 7 percentage point decrease in the Latino turnout disparity.

The biggest effects are to partisan affiliation disparities. Table 8 shows that a switch to more open primary rules reduces the unaffiliated share disparity between primary and general elections by over 10 percentage points, on average. It reduces the over-representation

[10]These regressions use 7 years of data instead of 8 years because the only regularly scheduled election L2 captures in 2017, for Virginia's state legislature and executive, has primary election data available but not general election data.

Table 7: **Opening Primaries Reduces Voter Turnout Gap Between Primary and General Elections**

	Voter Turnout Gap				
	Overall	Black	Asian	Latino	White
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	-0.079	-0.046	-0.042	-0.035	-0.094
	(0.014)	(0.013)	(0.008)	(0.008)	(0.018)
Open	-0.070	-0.055	-0.043	-0.042	-0.082
	(0.017)	(0.014)	(0.010)	(0.008)	(0.020)
Top-Four	-0.168	-0.120	-0.122	-0.071	-0.198
	(0.016)	(0.014)	(0.010)	(0.008)	(0.019)
Observations	1398	1398	1398	1398	1398
States	50	50	50	50	50
Years	7	7	7	7	7
Outcome Mean	0.309	0.207	0.251	0.182	0.337
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Outcome is the absolute value of the difference between primary and general electorate. The omitted primary type is closed. The data range is 2014–2024.

of Democrats and Republicans by a similar amount. In other words, after implementing more open primaries, the primary electorate looks more similar to the general electorate in terms of partisan representation. Table 9 shows there are also modest but detectable decreases in the compositional disparity for racial minorities. On average, the compositional gap in nonwhite electorate share decreases by half a percentage point when states switch to more open primary rules. This is shared roughly equally across racial groups.

Finally, Table 10 shows that opening up primaries significantly decreases gender, age, education, and veteran disparities between primary and general elections. The biggest effects are in age: a move from closed to open primaries reduces the median age gap by over 1.5 years, on average. Unaffiliated, third-party, racial minority, male, and young voters are all underrepresented in primary electorates, as shown in Table 1. Opening primaries to unaffiliated voters alters the primary electorate in ways that reduce representational disparities between who casts votes in primary elections and who is eligible to participate. While these

Table 8: **Effect of Opening Primaries on Partisan Composition Gap Between Primary and General Electorate**

	Partisan Share Gap			
	Unaffiliated	Third-party	Democratic	Republican
	(1)	(2)	(3)	(4)
Open To Unaffiliated	-0.100	0.002	-0.050	-0.060
	(0.021)	(0.001)	(0.009)	(0.018)
Open	-0.113	0.003	-0.051	-0.070
	(0.025)	(0.001)	(0.011)	(0.021)
Top-Four	-0.121	-0.001	-0.033	-0.110
	(0.023)	(0.001)	(0.011)	(0.021)
Observations	1398	1398	1398	1398
States	50	50	50	50
Years	7	7	7	7
Outcome Mean	0.122	0.010	0.073	0.093
State × Party × Office FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Outcome is the absolute value of the difference between primary and general electorate. The omitted primary type is closed. The data range is 2014–2024.

Table 9: **Effect of Opening Primaries on Racial Composition Gap Between Primary and General Electorate**

	Racial Share Gap				
	Black	Asian	Latino	Nonwhite	White
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	-0.001	-0.001	-0.002	-0.004	-0.004
	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Open	-0.000	-0.002	-0.002	-0.007	-0.007
	(0.001)	(0.000)	(0.002)	(0.002)	(0.002)
Top-Four	-0.004	-0.004	-0.012	-0.004	-0.004
	(0.001)	(0.001)	(0.002)	(0.003)	(0.003)
Observations	1398	1398	1398	1398	1398
States	50	50	50	50	50
Years	7	7	7	7	7
Outcome Mean	0.013	0.007	0.030	0.034	0.034
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Outcome is the absolute value of the difference between primary and general electorate. The omitted primary type is closed. The data range is 2014–2024.

tests rest on a relatively small set of state reforms, it is the strongest evidence to date that primary reforms can reduce participation disparities at the ballot box.

Table 10: **Effect of Opening Primaries on Demographic Composition Gap Between Primary and General Electorate**

	Demographic Gap				
	Male Share	Median Age	WC Share	Low-Edu Share	Veteran Share
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	-0.007	-1.671	0.000	-0.008	-0.006
	(0.002)	(0.387)	(0.002)	(0.003)	(0.002)
Open	-0.005	-1.534	-0.002	-0.004	-0.005
	(0.002)	(0.411)	(0.001)	(0.001)	(0.002)
Top-Four	-0.005	-3.737	-0.001	-0.002	-0.007
	(0.002)	(0.427)	(0.001)	(0.002)	(0.002)
Observations	1398	1398	1200	1200	1200
States	50	50	50	50	50
Years	7	7	6	6	6
Outcome Mean	0.011	6.025	0.014	0.009	0.012
State $\times$ Party $\times$ Office FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Outcome is the absolute value of the difference between primary and general electorate. The omitted primary type is closed. The data range is 2014–2024.

6 Party Registration and Ideological Extremism

I have shown evidence that opening primaries increases the representation of unaffiliated voters. Does this extend to ideologically moderate voters? This is the primary concern of prior scholarship and the discourse around primary reform: that it can provide a solution to increasing partisan polarization and lead to the election of more moderate candidates.

The voter file does not contain information on voters ideology. Therefore, I rely on 2018 CES survey data to examine the link between party registration and ideology, which uses registration data from the vendor Catalist. I filter the analysis to verified registered voters and group voters by party registration: Democrat, Republican, 3rd party,[11] and Unaffiliated (consisting of Decline to State, Independent, No Party Affiliation, and Unknown/No Party

[11] The following parties are included: Conservative Party, Constitution Party, Green Party, Libertarian, Other, Reform Party, Socialist Party, and Working Families Party

Registration). I measure ideological extremism by taking the absolute value of the distance between voter's self-identified ideology on a standard 5-point scale and the Moderate response (3). This takes values between 0 (Moderate) and 2 (Very liberal or Very conservative). I use the survey weights provided by the researchers so the data is representative of the pool of registered voters.

Figure A.7 in the online appendix shows descriptive means of average ideological extremism by party registration. 3rd party registrants tend to be the most ideologically moderate, followed by unaffiliated registrants, registered Democrats, and registered Republicans. I further examine the relationship between party registration and ideological extremism in Table 11. The reference category is unaffiliated voters. Column 1 shows the simple bivariate relationship illustrated in Figure rA.7. Column 2 introduces standard demographic controls and Column 3, my preferred specification, introduces state fixed effects.

Table 11: **Party Registration and Ideological Extremism**

	Ideological Extremism		
	(1)	(2)	(3)
Democrat	0.031	0.051	0.232
	(0.012)	(0.012)	(0.017)
Republican	0.156	0.129	0.311
	(0.013)	(0.013)	(0.018)
3rd Party	-0.157	-0.169	0.005
	(0.046)	(0.046)	(0.048)
Observations	38253	38245	38245
Outcome Mean	1.017	1.017	1.017
Demographic Controls	No	Yes	Yes
State FEs	No	No	Yes

Standard errors in parentheses. Reference category is Unaffiliated. Ideological extremism is measured as absolute distance from the ideological center on a 5-point ideology scale, with 0 indicating moderate, 1 indicating Liberal/Conservative, and 2 indicating Very Liberal/Very Conservative. Data from 2018 Cooperative Election Study filtered to verified registered voters. Regressions are run with weights provided by the survey team. Demographic controls are age, gender, education, race, and family income.

It is clear that unaffiliated registrants are substantially less ideologically extreme than registered Democrats and Republicans. The point estimates in column 3 suggest that registered Democrats are 23% more ideologically extreme than unaffiliated voters on average and registered Republicans are 31% more ideologically extreme (assuming equal distances on the ideology scale). There is no significant difference in the ideological extremity of unaffiliated and 3rd party registrants. The relationship holds when subsetting to verified primary voters, shown in Table A.20. While this does not prove that opening primaries leads to more moderate electorates, it is certainly consistent with that causal story.

7 Conclusion

It has been proposed that more open and nonpartisan primaries can help increase voter turnout and facilitate a more representative electorate when nominating candidates for the general election. Using the most detailed collection of state primary participation rules to date, a decade of nationwide voter file data, cutting-edge racial imputation techniques, and a difference-in-differences design, I find that primary electorates are less representative than general electorates and that opening primary elections increases turnout and the representativeness of the primary electorate. The evidence presented here supports these findings across a variety of demographic groups and other characteristics, including partisanship. Open primary rules lead to much higher participation from unaffiliated voters and shapes the electorate to be more representative of the pool of eligible voters. In other words, open and nonpartisan primaries do not just make it theoretically easier for unaffiliated voters to participate. Voters actually participate at higher rates and in a way that better reflects the general and eligible electorates. Given the strong relationship between party registration and ideology, it is likely that unaffiliated voter participation leads to more moderate electorates.

As policymakers and voters across the country consider adopting primary reforms, the findings of this analysis are clear: more open primaries will increase turnout and make primary voters more representative of the overall electorate. The obverse is likely true as well: states considering a return to closed primary rules will likely witness fewer and less representative voters participate in primary elections.

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Online Appendix

Intended for online publication only.

Contents

A.1 Changes in Primary Type By Party and Region, 2000–2024

Figure A.1 examines changes in congressional primary rules, with Democratic primaries on the left panel and Republican primaries on the right panel. The y-axis is the number of states holding Congressional primaries using the specified type. Republicans are more likely to use closed primaries, whereas Democrats are more likely to open their congressional primaries to unaffiliated voters. The decline in closed primary type and the increase in open to unaffiliated type has been quite dramatic on the Democratic party side, with the number of states allowing unaffiliated voters to participate on the Democratic side almost doubling over the past twenty years. 2024 is the first year when more states used an open to unaffiliated rule for their Democratic primary than a closed rule. Republican congressional primaries have also seen a decline in closed rules and a rise in open to unaffiliated rules, albeit to a lesser degree.

Figure A.1: **Type of Congressional Primaries by Party, 2000-2024**. This graph displays the percentage of primary elections with each primary type in even-year congressional elections between 2000 and 2024.

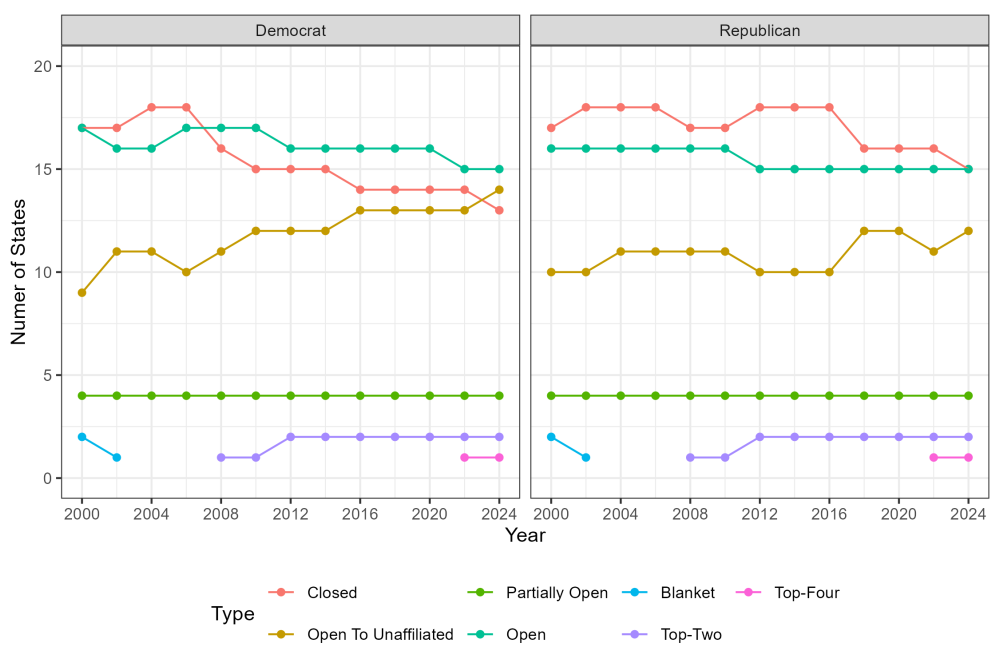
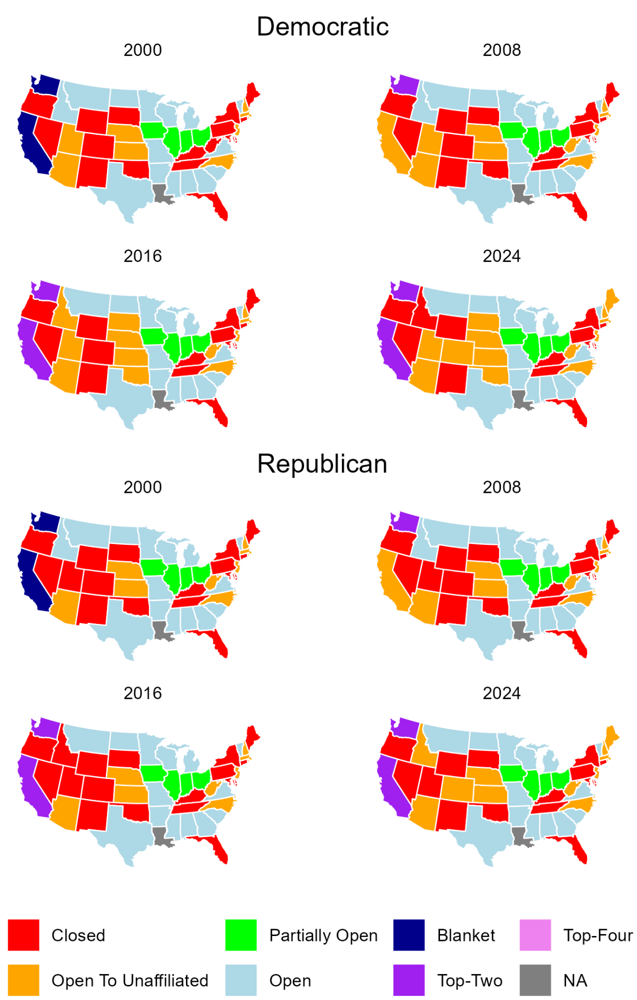


Figure A.2 maps each state's congressional primary type in four snapshots over the past quarter-century, with Democratic primary rules in the top panel and Republican primary rules in the bottom panel. For both major parties, it appears most of the reforms have occurred in New England and in the Mountain West states. Beyond these reforms, regional variation in primary type is also apparent. Most Southern states use open primaries, states in the Midwest tend to use partially open primaries, and New England and Mountain West states use a mix of closed and open to unaffiliated rules. Nonpartisan primaries are mostly confined to the Pacific West.

Figure A.2: **Map of Congressional Primary Types by Party, 2000-2024**. This graph displays a map of congressional primary type separately for the Democratic and Republican parties at four snapshots in time: 2000, 2008, 2016, and 2024.



A.2 L2 Voter File Dates Used

Following best practices of using voter file data detailed by Kim and Fraga (2022), this table lists the exact state file date used for every election in the analysis, listed by state. This is important for transparency because the composition of the file can change day-by-day. In general, the file closest to the election with recorded votes was used. I compared file dates recorded with this method that were within 2 months of the election date with all other file dates for that state, and used the file date with the most votes recorded for the election of interest. In most cases, this was the next file grab available, though in some cases this method yielded a file that was up to a year after the target election. This method tried to balance two competing priorities: ensuring I accurately capture all eligible votes recorded, and ensuring I accurately capture the list of voters that were registered at the time the election took place. Election grabs farther out may record more votes for idiosyncratic reasons with that state's voter records, but it also will be less reflective of the election's list of registered voters (with some voters leaving the file due to death or moving out of the state, and some voters entering the file due to coming of age or moving into the state).

Table A.1: L2 Voter File Dates Used, 2014–2024

State	Election Date	Election	File Date	State	Election Date	Election	File Date
AK	2014-08-19	Primary	2015-03-13	MT	2022-06-07	Primary	2022-12-01
AK	2014-11-04	General	2015-03-13	MT	2022-11-08	General	2023-05-18
AK	2016-08-16	Primary	2017-01-27	MT	2024-06-04	Primary	2024-10-04
AK	2016-11-08	General	2017-01-27	MT	2024-11-05	General	2025-06-23
AK	2018-08-21	Primary	2018-10-02	NC	2014-05-06	Primary	2015-07-29
AK	2018-11-06	General	2019-02-11	NC	2014-11-04	General	2015-07-29
AK	2020-08-18	Primary	2020-10-09	NC	2016-03-15	Primary	2016-05-26
AK	2020-11-03	General	2021-02-03	NC	2016-06-07	Primary	2016-10-19
AK	2022-08-16	Primary	2022-12-01	NC	2016-11-08	General	2017-01-12
AK	2022-11-08	General	2023-04-08	NC	2018-05-08	Primary	2018-06-28
AK	2024-08-20	Primary	2024-12-03	NC	2018-11-06	General	2019-02-01

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
AK	2024-11-05	General	2025-02-14	NC	2020-03-03	Primary	2021-01-28
AL	2014-06-03	Primary	2015-04-10	NC	2020-11-03	General	2021-01-28
AL	2014-11-04	General	2015-04-10	NC	2022-05-17	Primary	2022-08-23
AL	2016-03-01	Primary	2016-09-23	NC	2022-11-08	General	2023-02-15
AL	2016-11-08	General	2017-03-07	NC	2024-03-05	Primary	2024-06-26
AL	2018-06-05	Primary	2018-07-07	NC	2024-11-05	General	2025-01-10
AL	2018-11-06	General	2019-01-27	ND	2014-06-10	Primary	2015-04-15
AL	2020-03-03	Primary	2020-04-10	ND	2014-11-04	General	2015-04-15
AL	2020-11-03	General	2021-02-04	ND	2016-06-14	Primary	2016-09-28
AL	2022-05-24	Primary	2022-11-15	ND	2016-11-08	General	2017-02-09
AL	2022-11-08	General	2023-06-14	ND	2018-06-12	Primary	2018-09-08
AL	2024-03-05	Primary	2024-12-03	ND	2018-11-06	General	2019-03-22
AL	2024-11-05	General	2025-10-07	ND	2020-06-09	Primary	2020-10-01
AR	2014-05-20	Primary	2015-03-24	ND	2020-11-03	General	2021-03-18
AR	2014-11-04	General	2015-03-24	ND	2022-06-14	Primary	2022-10-14
AR	2016-03-01	Primary	2016-05-26	ND	2022-11-08	General	2023-09-19
AR	2016-11-08	General	2017-03-29	ND	2024-06-11	Primary	2025-02-07
AR	2018-05-22	Primary	2018-09-01	ND	2024-11-05	General	2025-07-09
AR	2018-11-06	General	2019-04-12	NE	2014-05-13	Primary	2015-03-25
AR	2020-03-03	Primary	2020-04-11	NE	2014-11-04	General	2015-03-25
AR	2020-11-03	General	2021-03-16	NE	2016-05-10	Primary	2016-07-02
AR	2022-05-24	Primary	2022-11-15	NE	2016-11-08	General	2017-01-13
AR	2022-11-08	General	2023-08-29	NE	2018-05-15	Primary	2018-07-11
AR	2024-03-05	Primary	2024-10-02	NE	2018-11-06	General	2019-01-10
AR	2024-11-05	General	2025-06-20	NE	2020-05-12	Primary	2020-10-01
AZ	2014-08-26	Primary	2015-04-22	NE	2020-11-03	General	2021-01-20
AZ	2014-11-04	General	2015-04-22	NE	2022-05-10	Primary	2022-10-04
AZ	2016-03-22	Primary	2016-10-03	NE	2022-11-08	General	2023-04-01
AZ	2016-08-30	Primary	2017-04-12	NE	2024-05-14	Primary	2024-12-10
AZ	2016-11-08	General	2017-04-12	NE	2024-11-05	General	2025-05-15
AZ	2018-08-28	Primary	2019-05-08	NH	2014-09-09	Primary	2015-03-20

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
AZ	2018-11-06	General	2019-05-08	NH	2014-11-04	General	2015-03-20
AZ	2020-03-17	Primary	2020-10-01	NH	2016-02-09	Primary	2016-06-27
AZ	2020-08-04	Primary	2020-10-23	NH	2016-09-13	Primary	2018-02-17
AZ	2020-11-03	General	2021-01-13	NH	2016-11-08	General	2018-02-17
AZ	2022-08-02	Primary	2022-10-08	NH	2018-09-11	Primary	2019-04-10
AZ	2022-11-08	General	2023-03-18	NH	2018-11-06	General	2019-04-10
AZ	2024-03-19	Primary	2024-06-26	NH	2020-02-11	Primary	2020-10-01
AZ	2024-07-30	Primary	2024-11-27	NH	2020-09-08	Primary	2021-03-25
AZ	2024-11-05	General	2025-03-04	NH	2020-11-03	General	2021-03-25
CA	2014-06-03	Primary	2015-01-29	NH	2022-09-13	Primary	2023-06-16
CA	2014-11-04	General	2015-01-29	NH	2022-11-08	General	2023-06-16
CA	2016-06-07	Primary	2016-09-29	NH	2024-01-23	Primary	2024-10-04
CA	2016-11-08	General	2017-03-25	NH	2024-09-10	Primary	2025-03-28
CA	2018-06-05	Primary	2018-08-17	NH	2024-11-05	General	2025-10-14
CA	2018-11-06	General	2019-01-31	NJ	2014-06-03	Primary	2015-02-25
CA	2020-03-03	Primary	2020-12-05	NJ	2014-11-04	General	2015-02-25
CA	2020-11-03	General	2020-12-05	NJ	2015-06-02	Primary	2016-05-23
CA	2022-06-07	Primary	2022-12-17	NJ	2015-11-03	General	2016-05-23
CA	2022-11-08	General	2022-12-17	NJ	2016-06-07	Primary	2016-09-29
CA	2024-03-05	Primary	2024-06-26	NJ	2016-11-08	General	2017-03-31
CA	2024-11-05	General	2025-01-10	NJ	2018-06-05	Primary	2018-10-16
CO	2014-06-24	Primary	2015-05-05	NJ	2018-11-06	General	2019-03-01
CO	2014-11-04	General	2015-05-05	NJ	2020-07-07	Primary	2020-10-01
CO	2016-03-01	Primary	2020-05-10	NJ	2020-11-03	General	2021-03-11
CO	2016-06-28	Primary	2016-10-13	NJ	2022-06-07	Primary	2022-08-22
CO	2016-11-08	General	2016-12-15	NJ	2022-11-08	General	2023-01-31
CO	2018-06-26	Primary	2018-08-08	NJ	2024-06-04	Primary	2025-01-15
CO	2018-11-06	General	2018-12-20	NJ	2024-11-05	General	2025-03-24
CO	2020-03-03	Primary	2020-05-10	NM	2014-06-03	Primary	2015-03-19
CO	2020-06-30	Primary	2021-05-28	NM	2014-11-04	General	2015-03-19
CO	2020-11-03	General	2021-05-28	NM	2016-06-07	Primary	2016-09-28

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
CO	2022-06-28	Primary	2022-12-07	NM	2016-11-08	General	2017-02-08
CO	2022-11-08	General	2023-01-14	NM	2018-06-05	Primary	2018-08-21
CO	2024-03-05	Primary	2024-06-26	NM	2018-11-06	General	2019-02-22
CO	2024-06-25	Primary	2025-01-10	NM	2020-06-02	Primary	2020-10-01
CO	2024-11-05	General	2025-01-15	NM	2020-11-03	General	2021-02-25
CT	2014-08-12	Primary	2015-03-25	NM	2022-06-07	Primary	2022-11-28
CT	2014-11-04	General	2015-03-25	NM	2022-11-08	General	2023-09-04
CT	2016-04-26	Primary	2017-01-20	NM	2024-06-04	Primary	2025-01-31
CT	2016-08-09	Primary	2017-01-20	NM	2024-11-05	General	2025-10-10
CT	2016-11-08	General	2017-01-20	NV	2014-06-10	Primary	2015-01-30
CT	2018-08-14	Primary	2019-03-06	NV	2014-11-04	General	2015-01-30
CT	2018-11-06	General	2019-03-06	NV	2016-06-14	Primary	2016-10-07
CT	2020-08-11	Primary	2021-03-30	NV	2016-11-08	General	2017-01-13
CT	2020-11-03	General	2021-03-30	NV	2018-06-12	Primary	2018-08-10
CT	2022-08-09	Primary	2022-12-14	NV	2018-11-06	General	2019-01-23
CT	2022-11-08	General	2023-05-24	NV	2020-06-09	Primary	2020-10-01
CT	2024-04-02	Primary	2025-02-20	NV	2020-11-03	General	2020-12-17
CT	2024-08-13	Primary	2025-07-31	NV	2022-06-14	Primary	2022-08-25
CT	2024-11-05	General	2025-07-31	NV	2022-11-08	General	2023-02-01
DC	2014-04-01	Primary	2015-03-07	NV	2024-02-06	Primary	2024-09-05
DC	2014-11-04	General	2015-03-07	NV	2024-06-11	Primary	2024-09-05
DC	2016-06-14	Primary	2016-09-23	NV	2024-11-05	General	2025-02-28
DC	2016-11-08	General	2017-02-15	NY	2014-06-24	Primary	2015-01-15
DC	2018-06-19	Primary	2019-01-17	NY	2014-09-09	Primary	2015-03-25
DC	2018-11-06	General	2019-01-17	NY	2014-11-04	General	2015-03-25
DC	2020-06-02	Primary	2020-10-01	NY	2015-09-10	Primary	2016-02-23
DC	2020-11-03	General	2021-01-30	NY	2015-11-03	General	2016-02-23
DC	2022-06-21	Primary	2022-09-21	NY	2016-04-19	Primary	2016-10-23
DC	2022-11-08	General	2023-09-18	NY	2016-06-28	Primary	2016-10-23
DC	2024-06-04	Primary	2025-01-10	NY	2016-09-13	Primary	2017-03-13
DC	2024-11-05	General	2025-06-04	NY	2016-11-08	General	2017-03-13

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
DE	2014-09-09	Primary	2015-02-23	NY	2017-09-12	Primary	2018-02-16
DE	2014-11-04	General	2015-02-23	NY	2017-11-07	General	2018-02-16
DE	2016-04-26	Primary	2016-06-18	NY	2018-06-26	Primary	2018-08-14
DE	2016-09-13	Primary	2017-01-17	NY	2018-09-13	Primary	2019-02-27
DE	2016-11-08	General	2017-01-17	NY	2018-11-06	General	2019-02-27
DE	2018-09-06	Primary	2019-04-02	NY	2020-06-23	Primary	2020-10-15
DE	2018-11-06	General	2019-04-02	NY	2020-11-03	General	2021-03-15
DE	2020-07-07	Primary	2020-10-23	NY	2022-06-28	Primary	2022-08-30
DE	2020-09-15	Primary	2021-03-24	NY	2022-11-08	General	2023-03-15
DE	2020-11-03	General	2021-03-24	NY	2024-04-02	Primary	2024-10-15
DE	2022-09-13	Primary	2022-10-26	NY	2024-06-25	Primary	2024-10-15
DE	2022-11-08	General	2023-05-20	NY	2024-11-05	General	2025-05-10
DE	2024-09-10	Primary	2025-05-10	OH	2014-05-06	Primary	2015-01-08
DE	2024-11-05	General	2025-05-10	OH	2014-11-04	General	2015-01-08
FL	2014-08-26	Primary	2015-01-28	OH	2015-05-05	Primary	2015-07-29
FL	2014-11-04	General	2016-06-14	OH	2015-11-03	General	2016-02-03
FL	2016-03-15	Primary	2016-06-14	OH	2016-03-15	Primary	2016-05-22
FL	2016-08-30	Primary	2017-01-27	OH	2016-11-08	General	2017-01-09
FL	2016-11-08	General	2017-01-27	OH	2017-05-02	Primary	2018-02-17
FL	2018-08-28	Primary	2019-02-08	OH	2017-11-07	General	2018-02-17
FL	2018-11-06	General	2019-02-08	OH	2018-05-08	Primary	2018-06-28
FL	2020-03-17	Primary	2020-10-01	OH	2018-11-06	General	2019-01-22
FL	2020-08-18	Primary	2021-02-04	OH	2020-04-28	Primary	2020-06-26
FL	2020-11-03	General	2021-02-04	OH	2020-11-03	General	2021-01-07
FL	2022-08-23	Primary	2023-02-09	OH	2022-08-02	Primary	2022-08-25
FL	2022-11-08	General	2023-02-09	OH	2022-11-08	General	2023-01-14
FL	2024-03-19	Primary	2024-06-26	OH	2024-03-19	Primary	2024-10-04
FL	2024-08-20	Primary	2024-10-18	OH	2024-11-05	General	2025-01-29
FL	2024-11-05	General	2025-02-11	OK	2014-06-24	Primary	2015-03-26
GA	2014-05-20	Primary	2015-05-16	OK	2014-11-04	General	2015-03-26
GA	2014-11-04	General	2015-05-16	OK	2016-03-01	Primary	2016-05-23

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
GA	2016-03-01	Primary	2016-07-08	OK	2016-06-28	Primary	2016-10-03
GA	2016-05-24	Primary	2016-07-08	OK	2016-11-08	General	2017-01-12
GA	2016-11-08	General	2017-01-26	OK	2018-06-26	Primary	2018-08-06
GA	2018-05-22	Primary	2018-07-05	OK	2018-11-06	General	2019-03-01
GA	2018-11-06	General	2020-07-24	OK	2020-03-03	Primary	2020-05-10
GA	2020-06-09	Primary	2020-07-24	OK	2020-06-30	Primary	2020-07-29
GA	2020-11-03	General	2022-08-27	OK	2020-11-03	General	2021-02-08
GA	2022-05-24	Primary	2022-08-27	OK	2022-06-28	Primary	2022-09-17
GA	2022-11-08	General	2022-12-05	OK	2022-11-08	General	2023-04-28
GA	2024-03-12	Primary	2024-06-26	OK	2024-03-05	Primary	2024-09-03
GA	2024-05-21	Primary	2024-09-05	OK	2024-06-18	Primary	2025-01-10
GA	2024-11-05	General	2024-12-24	OK	2024-11-05	General	2025-03-11
HI	2014-08-09	Primary	2015-03-05	OR	2014-05-20	Primary	2015-04-16
HI	2014-11-04	General	2015-03-05	OR	2014-11-04	General	2015-04-16
HI	2016-08-13	Primary	2017-03-22	OR	2016-05-17	Primary	2016-07-13
HI	2016-11-08	General	2017-03-22	OR	2016-11-08	General	2017-01-13
HI	2018-08-11	Primary	2019-04-05	OR	2018-05-15	Primary	2018-07-26
HI	2018-11-06	General	2019-04-05	OR	2018-11-06	General	2019-02-24
HI	2020-08-08	Primary	2020-10-22	OR	2020-05-19	Primary	2020-10-01
HI	2020-11-03	General	2021-04-01	OR	2020-11-03	General	2021-02-05
HI	2022-08-13	Primary	2022-10-25	OR	2022-05-17	Primary	2022-12-01
HI	2022-11-08	General	2023-09-12	OR	2022-11-08	General	2023-05-16
HI	2024-08-10	Primary	2025-01-10	OR	2024-05-21	Primary	2025-01-10
HI	2024-11-05	General	2025-07-21	OR	2024-11-05	General	2025-05-10
IA	2014-06-03	Primary	2015-01-27	PA	2014-05-20	Primary	2015-05-01
IA	2014-11-04	General	2015-01-27	PA	2014-11-04	General	2015-05-01
IA	2016-06-07	Primary	2016-10-18	PA	2015-05-19	Primary	2015-08-11
IA	2016-11-08	General	2017-01-31	PA	2015-11-03	General	2016-03-08
IA	2018-06-05	Primary	2018-08-25	PA	2016-04-26	Primary	2016-07-13
IA	2018-11-06	General	2019-03-06	PA	2016-11-08	General	2017-02-14

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
IA	2020-06-02	Primary	2020-10-01	PA	2017-05-16	Primary	2018-02-20
IA	2020-11-03	General	2021-03-04	PA	2017-11-07	General	2018-02-20
IA	2022-06-07	Primary	2022-10-18	PA	2018-05-15	Primary	2018-08-22
IA	2022-11-08	General	2023-04-06	PA	2018-11-06	General	2019-02-15
IA	2024-06-04	Primary	2024-10-03	PA	2020-06-02	Primary	2020-10-01
IA	2024-11-05	General	2025-04-24	PA	2020-11-03	General	2021-02-17
ID	2014-05-20	Primary	2015-02-23	PA	2022-05-17	Primary	2022-08-30
ID	2014-11-04	General	2015-02-23	PA	2022-11-08	General	2023-02-02
ID	2016-03-08	Primary	2016-05-31	PA	2024-04-23	Primary	2024-09-06
ID	2016-05-17	Primary	2016-10-05	PA	2024-11-05	General	2025-05-10
ID	2016-11-08	General	2017-03-20	RI	2014-09-09	Primary	2015-03-06
ID	2018-05-15	Primary	2018-08-21	RI	2014-11-04	General	2015-03-06
ID	2018-11-06	General	2019-03-04	RI	2016-04-26	Primary	2016-10-03
ID	2020-03-10	Primary	2020-08-14	RI	2016-09-13	Primary	2017-01-18
ID	2020-05-19	Primary	2020-10-04	RI	2016-11-08	General	2017-01-18
ID	2020-11-03	General	2021-03-16	RI	2018-09-12	Primary	2019-03-15
ID	2022-05-17	Primary	2022-10-15	RI	2018-11-06	General	2019-03-15
ID	2022-11-08	General	2023-09-18	RI	2020-06-02	Primary	2020-10-01
ID	2024-03-02	Primary	2025-02-11	RI	2020-09-08	Primary	2020-10-01
ID	2024-05-21	Primary	2025-07-17	RI	2020-11-03	General	2021-03-16
ID	2024-11-05	General	2025-07-17	RI	2022-09-13	Primary	2023-09-20
IL	2014-03-18	Primary	2015-01-08	RI	2022-11-08	General	2023-09-20
IL	2014-11-04	General	2015-03-02	RI	2024-04-02	Primary	2024-09-05
IL	2016-03-15	Primary	2016-09-23	RI	2024-09-10	Primary	2025-05-10
IL	2016-11-08	General	2017-03-17	RI	2024-11-05	General	2025-05-10
IL	2018-03-20	Primary	2018-07-28	SC	2014-06-10	Primary	2014-10-22
IL	2018-11-06	General	2019-02-21	SC	2014-11-04	General	2015-04-09
IL	2020-03-17	Primary	2020-10-01	SC	2016-02-27	Primary	2016-07-01
IL	2020-11-03	General	2021-03-05	SC	2016-06-14	Primary	2017-02-24
IL	2022-06-28	Primary	2022-10-19	SC	2016-11-08	General	2017-02-24

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
IL	2022-11-08	General	2023-03-18	SC	2018-06-12	Primary	2018-09-11
IL	2024-03-19	Primary	2024-09-03	SC	2018-11-06	General	2019-03-12
IL	2024-11-05	General	2025-05-10	SC	2020-02-29	Primary	2020-06-04
IN	2014-05-06	Primary	2015-05-06	SC	2020-06-09	Primary	2020-10-01
IN	2014-11-04	General	2015-05-06	SC	2020-11-03	General	2021-04-16
IN	2016-05-03	Primary	2016-09-23	SC	2022-06-14	Primary	2022-11-11
IN	2016-11-08	General	2017-04-07	SC	2022-11-08	General	2023-05-11
IN	2018-05-08	Primary	2018-09-01	SC	2024-02-24	Primary	2024-09-05
IN	2018-11-06	General	2019-02-13	SC	2024-06-11	Primary	2024-10-04
IN	2020-06-02	Primary	2020-10-01	SC	2024-11-05	General	2025-04-07
IN	2020-11-03	General	2021-01-15	SD	2014-06-03	Primary	2015-03-13
IN	2022-05-03	Primary	2022-10-08	SD	2014-11-04	General	2015-03-13
IN	2022-11-08	General	2023-03-27	SD	2016-06-07	Primary	2016-09-28
IN	2024-05-07	Primary	2025-02-18	SD	2016-11-08	General	2017-02-20
IN	2024-11-05	General	2025-10-02	SD	2018-06-05	Primary	2019-01-14
KS	2014-08-05	Primary	2015-02-26	SD	2018-11-06	General	2019-01-14
KS	2014-11-04	General	2015-02-26	SD	2020-06-02	Primary	2020-10-01
KS	2016-08-02	Primary	2017-02-16	SD	2020-11-03	General	2021-01-22
KS	2016-11-08	General	2017-02-16	SD	2022-06-07	Primary	2022-09-27
KS	2018-08-07	Primary	2019-01-31	SD	2022-11-08	General	2023-09-20
KS	2018-11-06	General	2019-01-31	SD	2024-06-04	Primary	2025-01-10
KS	2020-08-04	Primary	2021-03-16	SD	2024-11-05	General	2025-05-27
KS	2020-11-03	General	2021-03-16	TN	2014-08-07	Primary	2015-02-23
KS	2022-08-02	Primary	2022-11-28	TN	2014-11-04	General	2015-02-23
KS	2022-11-08	General	2023-09-18	TN	2016-03-01	Primary	2016-10-02
KS	2024-03-19	Primary	2024-10-03	TN	2016-08-04	Primary	2017-02-17
KS	2024-08-06	Primary	2024-10-22	TN	2016-11-08	General	2017-02-17
KS	2024-11-05	General	2025-08-13	TN	2018-08-02	Primary	2019-01-30
KY	2014-05-20	Primary	2015-03-05	TN	2018-11-06	General	2019-01-30
KY	2014-11-04	General	2015-03-05	TN	2020-03-03	Primary	2020-10-01

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
KY	2016-05-17	Primary	2016-09-23	TN	2020-08-06	Primary	2020-10-18
KY	2016-11-08	General	2017-03-03	TN	2020-11-03	General	2021-03-29
KY	2018-05-22	Primary	2018-09-29	TN	2022-08-04	Primary	2022-10-22
KY	2018-11-06	General	2019-05-02	TN	2022-11-08	General	2023-06-13
KY	2020-06-23	Primary	2020-10-01	TN	2024-03-05	Primary	2025-02-04
KY	2020-11-03	General	2021-05-11	TN	2024-08-01	Primary	2025-06-25
KY	2022-05-17	Primary	2022-10-11	TN	2024-11-05	General	2025-06-25
KY	2022-11-08	General	2023-09-06	TX	2014-03-04	Primary	2014-11-08
KY	2024-05-21	Primary	2025-04-10	TX	2014-11-04	General	2015-04-15
KY	2024-11-05	General	2025-10-20	TX	2016-03-01	Primary	2016-05-22
LA	2014-11-04	General	2015-02-23	TX	2016-11-08	General	2017-03-12
LA	2015-10-24	Primary	2016-01-29	TX	2018-03-06	Primary	2018-06-29
LA	2015-11-21	General	2016-01-29	TX	2018-11-06	General	2019-02-24
LA	2016-03-05	Primary	2016-06-09	TX	2020-03-03	Primary	2020-05-24
LA	2016-11-08	General	2017-02-14	TX	2020-11-03	General	2021-03-25
LA	2018-11-06	General	2019-01-15	TX	2022-03-01	Primary	2022-05-07
LA	2020-07-11	Primary	2020-10-18	TX	2022-11-08	General	2023-03-12
LA	2020-11-03	General	2021-01-22	TX	2024-03-05	Primary	2024-07-11
LA	2022-11-08	General	2023-06-20	TX	2024-11-05	General	2025-03-01
LA	2024-03-23	Primary	2024-10-03	UT	2014-06-24	Primary	2015-03-06
LA	2024-11-05	General	2024-12-16	UT	2014-11-04	General	2015-03-06
MA	2014-09-09	Primary	2015-04-02	UT	2016-06-28	Primary	2016-10-03
MA	2014-11-04	General	2015-04-02	UT	2016-11-08	General	2017-01-25
MA	2016-03-01	Primary	2016-09-28	UT	2018-06-26	Primary	2018-08-22
MA	2016-09-08	Primary	2017-04-11	UT	2018-11-06	General	2019-03-07
MA	2016-11-08	General	2017-04-11	UT	2020-03-03	Primary	2020-05-10
MA	2018-09-04	Primary	2019-01-18	UT	2020-06-30	Primary	2021-03-26
MA	2018-11-06	General	2019-02-14	UT	2020-11-03	General	2021-03-26
MA	2020-03-03	Primary	2020-05-29	UT	2022-06-28	Primary	2022-10-19
MA	2020-09-01	Primary	2021-01-19	UT	2022-11-08	General	2023-06-24

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
MA	2020-11-03	General	2021-01-19	UT	2024-03-05	Primary	2025-01-10
MA	2022-09-06	Primary	2023-09-12	UT	2024-06-25	Primary	2025-01-10
MA	2022-11-08	General	2023-09-12	UT	2024-11-05	General	2025-08-19
MA	2024-03-05	Primary	2024-10-04	VA	2014-06-10	Primary	2015-04-18
MA	2024-09-03	Primary	2025-02-10	VA	2014-11-04	General	2015-04-18
MA	2024-11-05	General	2025-07-03	VA	2015-06-09	Primary	2015-09-30
MD	2014-06-24	Primary	2015-02-25	VA	2015-11-03	General	2016-02-25
MD	2014-11-04	General	2015-02-25	VA	2016-03-01	Primary	2016-09-28
MD	2016-04-26	Primary	2016-10-03	VA	2016-06-14	Primary	2017-03-29
MD	2016-11-08	General	2017-01-20	VA	2016-11-08	General	2017-03-29
MD	2018-06-26	Primary	2018-12-14	VA	2017-06-13	Primary	2018-02-17
MD	2018-11-06	General	2018-12-14	VA	2018-06-12	Primary	2018-08-30
MD	2020-06-02	Primary	2020-10-01	VA	2018-11-06	General	2019-02-25
MD	2020-11-03	General	2021-02-15	VA	2020-03-03	Primary	2020-08-15
MD	2022-07-19	Primary	2022-10-19	VA	2020-06-23	Primary	2020-10-01
MD	2022-11-08	General	2023-09-10	VA	2020-11-03	General	2021-05-28
MD	2024-05-14	Primary	2025-01-10	VA	2022-06-21	Primary	2022-09-07
MD	2024-11-05	General	2025-06-04	VA	2022-11-08	General	2023-09-09
ME	2014-06-10	Primary	2015-04-29	VA	2024-03-05	Primary	2025-01-30
ME	2014-11-04	General	2015-04-29	VA	2024-06-18	Primary	2025-01-30
ME	2016-06-14	Primary	2016-10-05	VA	2024-11-05	General	2025-06-03
ME	2016-11-08	General	2017-04-07	VT	2014-08-26	Primary	2015-03-20
ME	2018-06-12	Primary	2018-09-26	VT	2014-11-04	General	2015-03-20
ME	2018-11-06	General	2019-07-17	VT	2016-03-01	Primary	2016-09-21
ME	2020-03-03	Primary	2020-06-18	VT	2016-08-09	Primary	2017-02-14
ME	2020-07-14	Primary	2020-10-01	VT	2016-11-08	General	2017-02-14
ME	2020-11-03	General	2021-05-28	VT	2018-08-14	Primary	2019-03-08
ME	2022-06-14	Primary	2023-06-07	VT	2018-11-06	General	2019-03-08
ME	2022-11-08	General	2023-06-07	VT	2020-03-03	Primary	2020-08-03
ME	2024-06-11	Primary	2025-02-05	VT	2020-08-11	Primary	2021-05-28

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
ME	2024-11-05	General	2025-10-08	VT	2020-11-03	General	2021-05-28
MI	2014-08-05	Primary	2015-02-28	VT	2022-08-09	Primary	2022-12-05
MI	2014-11-04	General	2015-02-28	VT	2022-11-08	General	2023-06-06
MI	2016-03-08	Primary	2017-02-21	VT	2024-03-05	Primary	2025-01-10
MI	2016-08-02	Primary	2017-02-21	VT	2024-08-13	Primary	2025-01-10
MI	2016-11-08	General	2017-02-21	VT	2024-11-05	General	2025-05-28
MI	2018-08-07	Primary	2019-03-22	WA	2014-08-05	Primary	2015-05-05
MI	2018-11-06	General	2019-03-22	WA	2014-11-04	General	2015-05-05
MI	2020-03-10	Primary	2020-08-14	WA	2015-08-04	Primary	2015-09-10
MI	2020-08-04	Primary	2020-10-01	WA	2015-11-03	General	2016-01-24
MI	2020-11-03	General	2021-01-30	WA	2016-05-24	Primary	2016-10-28
MI	2022-08-02	Primary	2022-10-06	WA	2016-08-02	Primary	2016-10-28
MI	2022-11-08	General	2023-04-21	WA	2016-11-08	General	2016-12-23
MI	2024-02-27	Primary	2024-06-26	WA	2017-08-01	Primary	2018-02-17
MI	2024-08-06	Primary	2024-10-02	WA	2017-11-07	General	2018-02-17
MI	2024-11-05	General	2025-05-10	WA	2018-08-07	Primary	2019-01-08
MN	2014-08-12	Primary	2015-03-03	WA	2018-11-06	General	2019-01-08
MN	2014-11-04	General	2015-03-03	WA	2020-03-10	Primary	2020-05-10
MN	2016-08-09	Primary	2017-03-10	WA	2020-08-04	Primary	2020-10-01
MN	2016-11-08	General	2017-03-10	WA	2020-11-03	General	2020-12-09
MN	2018-08-14	Primary	2019-04-02	WA	2022-08-02	Primary	2022-12-05
MN	2018-11-06	General	2019-04-02	WA	2022-11-08	General	2023-01-17
MN	2020-03-03	Primary	2020-10-01	WA	2024-03-12	Primary	2024-06-26
MN	2020-08-11	Primary	2020-10-19	WA	2024-08-06	Primary	2024-10-07
MN	2020-11-03	General	2021-02-14	WA	2024-11-05	General	2025-02-04
MN	2022-08-09	Primary	2022-10-27	WI	2014-08-12	Primary	2015-03-03
MN	2022-11-08	General	2023-09-12	WI	2014-11-04	General	2015-03-03
MN	2024-03-05	Primary	2024-10-03	WI	2016-04-05	Primary	2016-10-03
MN	2024-08-13	Primary	2025-02-14	WI	2016-08-09	Primary	2017-03-30
MN	2024-11-05	General	2025-05-14	WI	2016-11-08	General	2017-03-30

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
MO	2014-08-05	Primary	2015-03-02	WI	2018-08-14	Primary	2019-02-01
MO	2014-11-04	General	2015-03-02	WI	2018-11-06	General	2019-02-01
MO	2016-03-15	Primary	2016-07-01	WI	2020-04-07	Primary	2020-05-31
MO	2016-08-02	Primary	2017-02-08	WI	2020-08-11	Primary	2020-10-01
MO	2016-11-08	General	2017-02-08	WI	2020-11-03	General	2021-02-24
MO	2018-08-07	Primary	2018-10-05	WI	2022-08-09	Primary	2022-12-14
MO	2018-11-06	General	2019-02-16	WI	2022-11-08	General	2023-04-28
MO	2020-03-10	Primary	2020-06-23	WI	2024-04-02	Primary	2024-10-02
MO	2020-08-04	Primary	2021-02-11	WI	2024-11-05	General	2025-05-16
MO	2020-11-03	General	2021-02-11	WV	2014-05-13	Primary	2015-03-16
MO	2022-08-02	Primary	2022-11-19	WV	2014-11-04	General	2015-03-16
MO	2022-11-08	General	2023-05-10	WV	2016-05-10	Primary	2016-09-28
MO	2024-08-06	Primary	2025-02-01	WV	2016-11-08	General	2017-04-03
MO	2024-11-05	General	2025-05-20	WV	2018-05-08	Primary	2018-08-14
MS	2014-06-03	Primary	2015-03-17	WV	2018-11-06	General	2019-03-22
MS	2014-11-04	General	2015-03-17	WV	2020-06-09	Primary	2020-10-06
MS	2016-03-08	Primary	2016-10-03	WV	2020-11-03	General	2021-03-11
MS	2016-11-08	General	2017-03-07	WV	2022-05-10	Primary	2023-09-18
MS	2018-06-05	Primary	2018-09-18	WV	2022-11-08	General	2023-09-18
MS	2018-11-06	General	2019-03-11	WV	2024-05-14	Primary	2025-01-10
MS	2020-03-10	Primary	2020-10-01	WV	2024-11-05	General	2025-05-10
MS	2020-11-03	General	2021-03-23	WY	2014-08-19	Primary	2015-03-30
MS	2022-06-07	Primary	2022-10-27	WY	2014-11-04	General	2015-03-30
MS	2022-11-08	General	2023-09-19	WY	2016-08-16	Primary	2017-02-02
MS	2024-08-06	Primary	2025-01-10	WY	2016-11-08	General	2017-02-02
MS	2024-11-05	General	2025-06-10	WY	2018-08-21	Primary	2019-04-02
MT	2014-06-03	Primary	2015-03-27	WY	2018-11-06	General	2019-04-02
MT	2014-11-04	General	2015-03-27	WY	2020-08-18	Primary	2020-10-09
MT	2016-06-07	Primary	2016-10-03	WY	2020-11-03	General	2021-01-13
MT	2016-11-08	General	2017-01-25	WY	2022-08-16	Primary	2022-10-21

Table A.1: L2 Voter File Dates Used, 2014–2024 *(continued)*

State	Election Date	Election	File Date	State	Election Date	Election	File Date
MT	2018-06-05	Primary	2018-08-03	WY	2022-11-08	General	2023-09-19
MT	2018-11-06	General	2019-02-07	WY	2024-08-20	Primary	2025-07-09
MT	2020-06-02	Primary	2020-10-01	WY	2024-11-05	General	2025-07-09
MT	2020-11-03	General	2020-12-14				

Each row shows the L2 voter file snapshot used for a given state-election.

A.3 Additional Descriptive Tables

Table A.2 displays the average difference in turnout and composition of state primary elections compared with same state-year general elections across years divisible by two. In states with multiple primary elections in the same year, the characteristics of the first primary election is used. As in the Difference column in Table 1 in the main analysis, the calculation is general election characteristic − primary election characteristic. Positive values indicate a larger turnout/electorate share in primary elections vs. general elections, and negative values indicate a smaller turnout/electorate share in primaries.

The table shows broadly similar patterns to the composite results. Across all federal election years, primary election turnout is much lower than general election turnout. This ranges from a gap of 23 percentage points in 2014 to a gap of 45 percentage points in 2024. Race-specific turnout is also lower in primary elections across years. Primary electorate are proportionally whiter, older, more major party-affiliated, and less working class in every federal election cycle studied.

There is no clear evidence for over-time trends. In other words, the gap in turnout and electorate composition between primary and general elections does not seem to be growing systematically greater or smaller. One trend is clear: primary electorates least resemble general electorates in presidential election years. This is almost certainly due to the higher turnout of low-propensity voters when the president is elected, which does not proportionally trickle down to the primary cycle. For instance, the general-primary turnout gap is 23, 29, and 25 percentage points in the three midterm cycles studied, but 33, 40, and 45 percentage points in the three presidential cycles studied. Unaffiliated voters are underrepresented by 10–11 percentage points in midterm cycles, but by 14–15 percentage points in presidential cycles. There may also be small representational differences by cycle in education and class.

Table A.3 reports primary-general electorate differences by state. It again shows that primary elections are consistently different from general elections in terms of both the number

Table A.2: Differences in Turnout and Composition Between Primary and General Electorates By Year, 2014–2024

	2014	2016	2018	2020	2022	2024
Turnout						
Turnout	-23%	-33%	-29%	-40%	-25%	-45%
Black turnout	-15%	-25%	-21%	-29%	-13%	-32%
Latino turnout	-10%	-23%	-18%	-27%	-11%	-28%
Asian turnout	-13%	-29%	-24%	-38%	-18%	-38%
White turnout	-25%	-35%	-32%	-43%	-29%	-50%
Racial Demographics						
Share Nonwhite	-2%	-3%	-3%	-2%	-2%	-6%
Share Black	-1%	-1%	-1%	1%	0%	-1%
Share Latino	0%	-1%	-1%	-1%	-1%	-3%
Share Asian	0%	-1%	0%	-1%	0%	-1%
Share Other	0%	-1%	-1%	-1%	-1%	-1%
Share White	2%	3%	3%	2%	2%	6%
Other Demographics						
Median age of voters	4	5	5	5	4	8
Share Democratic	1%	6%	6%	14%	3%	5%
Share Republican	11%	10%	6%	1%	8%	12%
Share 3rd Party	-1%	-1%	-1%	-1%	-1%	-1%
Share Unaffiliated	-11%	-15%	-11%	-14%	-10%	-15%
Share Female	0%	0%	0%	2%	1%	1%
Share under 50k income	2%	0%	-	-	-	-
Share under 100k income	1%	0%	-	-	-	-
Share over 250k income	0%	0%	-	-	-	-
Share some college	-1%	1%	0%	1%	0%	-
Share working-class	-1%	-1%	-1%	-2%	-1%	-
Share veteran	1%	0%	1%	1%	1%	-

Each cell shows the average state primary turnout/electorate composition in that year minus the average state general election turnout/electorate composition. For share rows, positive values indicate over-representation in the primary election relative to the general election and negative values indicate under-representation. All data is averaged at the state level.

of voters that show up and the partisan and demographic makeup of the voters that do show up.

Table A.3: Differences in Turnout and Composition Between Primary and General Electorates By State, 2014–2024

	AK	AL	AR	AZ	CA	CO	CT	DC	DE	FL	GA	HI	IA	ID	IL	IN	KS	KY	LA	MA	MD	ME	MI	MN	MO	MS
Turnout																										
Turnout	-31%	-23%	-26%	-33%	-22%	-39%	-46%	-30%	-39%	-38%	-33%	-14%	-48%	-34%	-31%	-28%	-30%	-30%	-29%	-34%	-32%	-47%	-33%	-51%	-27%	-27%
Black turnout	-20%	-23%	-22%	-21%	-17%	-25%	-25%	-24%	-26%	-30%	-31%	-10%	-23%	-34%	-21%	-19%	-18%	-19%	-22%	-20%	-24%	-28%	-26%	-29%	-18%	-27%
Latino turnout	-17%	-12%	-15%	-23%	-22%	-23%	-28%	-24%	-22%	-36%	-25%	-10%	-22%	-19%	-23%	-16%	-16%	-14%	-16%	-27%	-28%	-14%	-19%	-26%	-12%	-9%
Asian turnout	-17%	-20%	-18%	-30%	-21%	-32%	-39%	-28%	-36%	-34%	-36%	-11%	-33%	-17%	-29%	-24%	-24%	-26%	-20%	-31%	-31%	-26%	-37%	-39%	-24%	-13%
White turnout	-37%	-24%	-27%	-38%	-22%	-44%	-52%	-36%	-45%	-40%	-35%	-21%	-50%	-36%	-35%	-30%	-33%	-31%	-33%	-35%	-37%	-48%	-35%	-54%	-29%	-27%
Racial Demographics																										
Share Nonwhite	-4%	-2%	-3%	-5%	-8%	-4%	-4%	-5%	0%	-9%	-6%	3%	-2%	-3%	-1%	-2%	-3%	-1%	4%	-4%	-1%	-1%	-3%	-2%	-1%	-3%
Share Black	-1%	-1%	-1%	-1%	-1%	0%	1%	-2%	2%	-1%	-2%	0%	-1%	0%	2%	0%	-1%	0%	5%	0%	3%	0%	-1%	-1%	0%	-3%
Share Latino	-1%	0%	-1%	-3%	-5%	-2%	-3%	-1%	-1%	-5%	-1%	-1%	-1%	-2%	-1%	-1%	-1%	0%	0%	-2%	-1%	0%	-1%	-1%	0%	0%
Share Asian	-1%	0%	0%	-1%	-1%	-1%	-1%	-1%	-1%	-1%	-1%	3%	0%	0%	-1%	0%	-1%	0%	0%	-1%	-1%	0%	-1%	-1%	0%	0%
Share Other	-1%	0%	-1%	0%	-1%	-1%	-1%	-1%	-1%	-1%	-1%	0%	0%	-1%	0%	0%	-1%	0%	-1%	-1%	-1%	-1%	-1%	0%	0%	0%
Share White	4%	2%	3%	5%	8%	4%	4%	5%	0%	9%	6%	-3%	2%	3%	1%	2%	3%	1%	-4%	4%	1%	1%	3%	2%	1%	3%
Other Demographics																										
Median age of voters	3	5	5	7	5	8	6	3	5	7	7	2	9	5	4	5	6	4	5	5	4	6	6	7	4	4
Share Democratic	2%	-3%	4%	10%	3%	5%	18%	14%	17%	2%	4%	1%	12%	-1%	13%	4%	1%	6%	6%	7%	13%	13%	2%	9%	0%	1%
Share Republican	3%	6%	18%	9%	3%	10%	19%	-3%	4%	14%	18%	1%	15%	23%	11%	19%	14%	0%	6%	-1%	1%	11%	6%	-3%	3%	20%
Share 3rd Party	-1%	0%	0%	-4%	-1%	-1%	-1%	-1%	-2%	-1%	0%	0%	-1%	-1%	0%	0%	-1%	-5%	0%	0%	-1%	-3%	0%	0%	0%	0%
Share Unaffiliated	-4%	-3%	-23%	-14%	-5%	-14%	-36%	-11%	-19%	-15%	-23%	-2%	-27%	-21%	-24%	-23%	-15%	0%	-11%	-6%	-12%	-21%	-7%	-6%	-4%	-21%
Share Female	1%	-1%	-1%	1%	0%	2%	1%	2%	2%	1%	0%	0%	1%	0%	-1%	0%	1%	1%	2%	0%	2%	1%	1%	1%	0%	-1%
Share under 50k income	0%	0%	0%	2%	1%	3%	1%	-1%	2%	1%	0%	1%	2%	0%	2%	1%	3%	2%	0%	1%	1%	0%	1%	5%	1%	-2%
Share under 100k income	-2%	0%	0%	3%	1%	3%	0%	0%	1%	-4%	0%	0%	1%	0%	-1%	1%	2%	1%	0%	1%	1%	-1%	0%	3%	1%	-1%
Share over 250k income	0%	0%	0%	0%	0%	0%	0%	0%	0%	1%	0%	0%	0%	0%	1%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Share some college	0%	0%	1%	0%	1%	-1%	1%	1%	0%	0%	0%	1%	0%	0%	0%	-1%	0%	-1%	1%	1%	0%	3%	0%	0%	0%	0%
Share working-class	-1%	0%	-1%	-1%	-1%	-1%	-3%	0%	-2%	-1%	-1%	-1%	-3%	-2%	-1%	-1%	-1%	-1%	-1%	-2%	-2%	-4%	-2%	-3%	-1%	-1%
Share veteran	1%	1%	1%	1%	1%	2%	1%	0%	0%	1%	1%	1%	2%	1%	0%	1%	1%	0%	0%	0%	0%	1%	1%	1%	1%	1%

Each cell shows the average state primary minus average general electorate composition across all elections. Continued in next table.

Table A.3: Differences in Turnout and Composition Between Primary and General Electorates By State, 2014–2024 (cont.)

	MT	NC	ND	NE	NH	NJ	NM	NV	NY	OH	OK	OR	PA	RI	SC	SD	TN	TX	UT	VA	VT	WA	WI	WV	WY
Turnout																									
Turnout	-27%	-37%	-33%	-30%	-30%	-36%	-30%	-36%	-33%	-31%	-23%	-32%	-27%	-36%	-31%	-34%	-26%	-30%	-34%	-40%	-38%	-25%	-36%	-19%	-22%
Black turnout	-15%	-32%	-26%	-21%	-16%	-21%	-19%	-24%	-20%	-23%	-19%	-18%	-19%	-12%	-28%	-11%	-20%	-26%	-21%	-30%	-18%	-15%	-26%	-13%	-16%
Latino turnout	-9%	-26%	-12%	-19%	-19%	-28%	-21%	-32%	-22%	-14%	-11%	-23%	-18%	-29%	-16%	-10%	-15%	-22%	-18%	-27%	-13%	-15%	-25%	-10%	-10%
Asian turnout	-10%	-38%	-22%	-24%	-27%	-36%	-28%	-24%	-25%	-32%	-17%	-27%	-31%	-21%	-21%	-16%	-21%	-34%	-22%	-39%	-29%	-21%	-32%	-17%	-10%
White turnout	-28%	-39%	-35%	-32%	-30%	-41%	-41%	-42%	-39%	-33%	-27%	-33%	-28%	-39%	-33%	-37%	-27%	-35%	-37%	-44%	-39%	-27%	-37%	-20%	-23%
Racial Demographics																									
Share Nonwhite	-2%	-4%	-2%	-3%	-2%	1%	-2%	-11%	7%	-3%	-4%	-4%	-1%	-1%	0%	-2%	-2%	-8%	-2%	-6%	-1%	-3%	-3%	-1%	-2%
Share Black	0%	-2%	-1%	-1%	0%	3%	0%	-1%	6%	-2%	-1%	0%	1%	0%	1%	0%	-1%	-2%	0%	-3%	0%	0%	-1%	0%	0%
Share Latino	0%	-1%	0%	-1%	0%	0%	-1%	-7%	2%	0%	-1%	-2%	-1%	-1%	-1%	0%	-1%	-3%	-1%	-1%	0%	-1%	-1%	0%	-1%
Share Asian	0%	-1%	0%	0%	0%	-2%	0%	-1%	-1%	-1%	0%	-1%	-1%	0%	0%	0%	0%	-2%	0%	-2%	0%	-1%	-1%	0%	0%
Share Other	-1%	0%	-1%	-1%	-1%	0%	-1%	-2%	0%	0%	-2%	-1%	0%	0%	0%	-1%	-1%	-1%	0%	0%	-1%	-1%	0%	0%	-1%
Share White	2%	4%	2%	3%	2%	-1%	2%	11%	-7%	3%	4%	4%	1%	1%	0%	2%	2%	8%	2%	6%	1%	3%	3%	1%	2%
Other Demographics																									
Median age of voters	6	6	4	6	3	5	6	9	5	5	4	6	5	4	5	5	4	7	7	5	5	4	4	3	2
Share Democratic	1%	2%	-1%	-2%	2%	18%	14%	7%	23%	5%	-1%	7%	8%	20%	2%	-4%	8%	-4%	11%	1%	8%	5%	3%	4%	0%
Share Republican	5%	2%	10%	10%	7%	9%	2%	7%	-2%	11%	6%	5%	2%	2%	4%	15%	17%	13%	13%	12%	0%	3%	2%	-1%	8%
Share 3rd Party	0%	0%	0%	-1%	0%	0%	-2%	-3%	-4%	0%	0%	-2%	-2%	0%	0%	-7%	0%	0%	-3%	0%	0%	0%	0%	0%	-1%
Share Unaffiliated	-6%	-4%	-9%	-8%	-9%	-27%	-14%	-11%	-17%	-16%	-5%	-10%	-7%	-22%	-6%	-3%	-25%	-9%	-21%	-13%	-8%	-9%	-5%	-3%	-7%
Share Female	2%	0%	1%	0%	0%	2%	2%	1%	3%	0%	0%	1%	1%	1%	0%	0%	-1%	-1%	3%	0%	1%	1%	0%	0%	1%
Share under 50k income	1%	0%	2%	1%	1%	5%	3%	1%	9%	0%	0%	1%	3%	0%	-2%	4%	1%	3%	2%	-3%	1%	2%	1%	1%	0%
Share under 100k income	0%	0%	0%	2%	0%	6%	1%	-1%	5%	1%	0%	0%	3%	0%	-4%	2%	1%	0%	1%	-5%	0%	2%	1%	1%	0%
Share over 250k income	0%	0%	0%	0%	0%	-1%	0%	0%	0%	0%	0%	0%	0%	0%	1%	0%	0%	1%	0%	1%	0%	0%	0%	0%	0%
Share some college	-1%	0%	-1%	-1%	0%	-2%	-1%	1%	-1%	0%	0%	0%	-1%	1%	0%	0%	-1%	-1%	-1%	3%	1%	0%	1%	0%	0%
Share working-class	-1%	-1%	-2%	-1%	-2%	0%	0%	-2%	-2%	-1%	-1%	-1%	-1%	-2%	-1%	-2%	0%	-1%	-2%	-3%	-3%	-1%	-3%	-1%	-1%
Share veteran	1%	1%	1%	1%	1%	1%	0%	1%	0%	1%	1%	1%	1%	0%	1%	1%	1%	1%	1%	1%	0%	1%	1%	0%	1%

Each cell shows the average state primary minus average general electorate composition across all elections.

A.4 Turnout and Compositional Effects - Always Closed or Switch to Open to Unaffiliated Sample

The regressions in the main analysis included multiple different primary reforms, including to nonpartisan primaries and between open and closed systems. I simplify the analysis here to only the most numerous change: from closed to open-to-unaffiliated primaries. To ensure a clean counterfactual comparison, I subset to state-party-offices that either always use a closed rule between 2014 and 2024 (the control group) or switch from closed to open-to-unaffiliated in that time span (the treated group). Tables A.4, A.5, A.6, and A.7 show the results and correspond to Tables 3, 4, 5, and 6 in the main analysis, respectively.

The results are broadly in line with the full specifications. Column 1 of Table A.4 shows that opening primaries to unaffiliated voters boosts turnout by 6.4 percentage points (compared to 4.9 percentage points estimated in column 1 of Table 3). It boosts the unaffiliated share of the electorate by 12 percentage points (compared with 13.8 percentage points), the nonwhite share by .4 percentage points (the same as the main effect estimated), and reduces the median age by 1.2 years (compared with 1.5 years). In short, the estimates are broadly consistent with this specification.

Because this specification removes potentially problematic comparisons between treatment and control primary types and produces similar point estimates to the main specification, I conduct randomization inference on both specifications in Figure 3 to show the validity of my findings. I use the full specification in the main analysis because it allows me to compare the effects of different primary reforms.

Table A.4: **Opening Primaries Increases Voter Turnout (Closed States Only)**

	Voter Turnout				
	Overall	Black	Asian	Latino	White
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	0.064	0.019	0.036	0.026	0.074
	(0.014)	(0.007)	(0.010)	(0.006)	(0.016)
Observations	595	595	595	595	595
States	22	22	22	22	22
Years	6	6	6	6	6
Outcome Mean	0.199	0.105	0.070	0.062	0.233
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Sample restricted to state-party-offices that switched from Closed to Open To Unaffiliated and those that remained Closed throughout. The omitted primary type is Closed. The data range is 2014–2024.

Table A.5: **Effect of Opening Primaries on Partisan Composition of Electorate (Closed States Only)**

	Partisan Share			
	Unaffiliated	Third-party	Democratic	Republican
	(1)	(2)	(3)	(4)
Open To Unaffiliated	0.159	-0.002	-0.053	-0.105
	(0.033)	(0.001)	(0.015)	(0.034)
Observations	595	595	595	595
States	22	22	22	22
Years	6	6	6	6
Outcome Mean	0.053	0.011	0.461	0.475
State × Party × Office FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Sample restricted to state-party-offices that switched from Closed to Open To Unaffiliated and those that remained Closed throughout. The omitted primary type is Closed. The data range is 2014–2024.

Table A.6: **Effect of Opening Primaries on Racial Composition of Electorate (Closed States Only)**

	Racial Share				
	Black	Asian	Latino	Nonwhite	White
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	-0.001	-0.001	0.007	0.009	-0.009
	(0.002)	(0.001)	(0.002)	(0.003)	(0.003)
Observations	595	595	595	595	595
States	22	22	22	22	22
Years	6	6	6	6	6
Outcome Mean	0.073	0.011	0.048	0.146	0.854
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Sample restricted to state-party-offices that switched from Closed to Open To Unaffiliated and those that remained Closed throughout. The omitted primary type is Closed. The data range is 2014–2024.

Table A.7: **Effect of Opening Primaries on Demographic Composition of Electorate (Closed States Only)**

	Demographic Share					
	Male Share	Median Age	Low-Income Share	WC Share	Low-Edu Share	Veteran Share
	(1)	(2)	(3)	(4)	(5)	(6)
Open To Unaffiliated	0.015	-1.825	0.006	-0.000	0.000	-0.004
	(0.002)	(0.481)	(0.005)	(0.001)	(0.003)	(0.003)
Observations	595	595	196	492	492	492
States	22	22	22	22	22	22
Years	6	6	2	5	5	5
Outcome Mean	0.455	61.971	0.288	0.270	0.681	0.057
State × Party × Office FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state-party-office unit in parentheses. Sample restricted to state-party-offices that switched from Closed to Open To Unaffiliated and those that remained Closed throughout. The omitted primary type is Closed. The data range is 2014–2024.

A.5 Turnout and Compositional Effects - Congressional Democrat Type

The regressions in the main analysis were at the state-office-party-election date level, and thus included state-office-party fixed effects. Since electorate characteristics were measured at the state-election date level, regressions measured a weighted average of the effect of primary type by the number of elections that took place (Democratic and Republican presidential, congressional, state executive, and state legislative offices).

This and the follow section simplify the analysis to the state-election level by subsetting to only Congressional Democratic and Congressional Republican offices, respectively. Therefore, I substitute state-office-party fixed effects for state fixed effects. The results are consistent with those in the main analysis. Louisiana is dropped from this analysis, since all of its Congressional primaries in this period have used the jungle system on the general election date in other states.

Table A.8: **Opening Primaries Increases Voter Turnout (Congressional Democratic Type)**

	Voter Turnout				
	Overall	Black	Asian	Latino	White
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	0.031	0.023	0.015	0.014	0.036
	(0.014)	(0.007)	(0.010)	(0.005)	(0.017)
Open	-0.058	-0.007	-0.014	-0.027	-0.086
	(0.007)	(0.005)	(0.004)	(0.005)	(0.008)
Observations	229	229	229	229	229
States	49	49	49	49	49
Years	6	6	6	6	6
Outcome Mean	0.210	0.103	0.076	0.061	0.245
State FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table A.9: **Effect of Opening Primaries on Partisan Composition of Electorate (Congressional Democratic Type)**

	Partisan Share			
	Unaffiliated	Third-party	Democratic	Republican
	(1)	(2)	(3)	(4)
Open To Unaffiliated	0.120	-0.004	-0.057	-0.060
	(0.049)	(0.003)	(0.024)	(0.049)
Open	-0.016	-0.005	-0.053	0.074
	(0.006)	(0.001)	(0.010)	(0.010)
Observations	229	229	229	229
States	49	49	49	49
Years	6	6	6	6
Outcome Mean	0.098	0.009	0.405	0.488
State FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table A.10: **Effect of Opening Primaries on Racial Composition of Electorate (Congressional Democratic Type)**

	Racial Share				
	Black	Asian	Latino	Nonwhite	White
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	0.001	-0.000	0.002	0.004	-0.004
	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)
Open	-0.006	0.001	-0.004	0.017	-0.017
	(0.003)	(0.001)	(0.001)	(0.003)	(0.003)
Observations	229	229	229	229	229
States	49	49	49	49	49
Years	6	6	6	6	6
Outcome Mean	0.069	0.022	0.036	0.146	0.854
State FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table A.11: **Effect of Opening Primaries on Demographic Composition of Electorate (Congressional Democratic Type)**

	Male Share	Median Age	Demographic Share WC Share	Demographic Share Low-Edu Share	Veteran Share
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	0.008	-1.152	0.001	-0.001	-0.007
	(0.005)	(0.331)	(0.004)	(0.008)	(0.001)
Open	0.006	-0.141	0.003	0.000	-0.003
	(0.001)	(0.255)	(0.001)	(0.001)	(0.001)
Observations	229	229	202	202	202
States	49	49	49	49	49
Years	6	6	5	5	5
Outcome Mean	0.464	61.712	0.289	0.680	0.057
State FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

A.6 Turnout and Compositional Effects - Congressional Republican Type

Table A.12: **Opening Primaries Increases Voter Turnout (Congressional Republican Type)**

	Voter Turnout				
	Overall (1)	Black (2)	Asian (3)	Latino (4)	White (5)
Open To Unaffiliated	0.038	-0.003	0.016	0.014	0.039
	(0.014)	(0.015)	(0.014)	(0.006)	(0.020)
Top-Four	0.097	0.004	0.030	0.042	0.125
	(0.018)	(0.017)	(0.015)	(0.008)	(0.023)
Observations	229	229	229	229	229
States	49	49	49	49	49
Years	6	6	6	6	6
Outcome Mean	0.210	0.103	0.076	0.061	0.245
State FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table A.13: **Effect of Opening Primaries on Partisan Composition of Electorate (Congressional Republican Type)**

	Partisan Share			
	Unaffiliated (1)	Third-party (2)	Democratic (3)	Republican (4)
Open To Unaffiliated	0.126	-0.004	-0.060	-0.062
	(0.070)	(0.003)	(0.023)	(0.067)
Top-Four	0.143	0.000	-0.007	-0.137
	(0.070)	(0.003)	(0.029)	(0.069)
Observations	229	229	229	229
States	49	49	49	49
Years	6	6	6	6
Outcome Mean	0.098	0.009	0.405	0.488
State FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table A.14: **Effect of Opening Primaries on Racial Composition of Electorate (Congressional Republican Type)**

	Racial Share				
	Black	Asian	Latino	Nonwhite	White
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	0.002	-0.001	0.003	0.005	-0.005
	(0.003)	(0.001)	(0.002)	(0.004)	(0.004)
Top-Four	0.008	-0.002	0.007	-0.012	0.012
	(0.005)	(0.002)	(0.003)	(0.006)	(0.006)
Observations	229	229	229	229	229
States	49	49	49	49	49
Years	6	6	6	6	6
Outcome Mean	0.069	0.022	0.036	0.146	0.854
State FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

Table A.15: **Effect of Opening Primaries on Demographic Composition of Electorate (Congressional Republican Type)**

	Demographic Share				
	Male Share	Median Age	WC Share	Low-Edu Share	Veteran Share
	(1)	(2)	(3)	(4)	(5)
Open To Unaffiliated	0.013	-0.736	-0.006	-0.002	0.001
	(0.002)	(0.603)	(0.002)	(0.007)	(0.007)
Top-Four	0.007	-0.589	-0.009	-0.003	0.004
	(0.003)	(0.712)	(0.003)	(0.007)	(0.007)
Observations	229	229	202	202	202
States	49	49	49	49	49
Years	6	6	5	5	5
Outcome Mean	0.464	61.712	0.289	0.680	0.057
State FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes

Robust standard errors clustered by state in parentheses. The omitted primary type is closed. The data range is 2014–2024.

A.7 Heterogeneous-Robust Estimators

This section reports the results of the Callaway and Sant'Anna (2021) dynamic difference-in-difference estimator. This removes potential bias in staggered treatment DID designs due to heterogeneous effects across units or time (Callaway and Sant'Anna 2021; de Chaisemartin and D'Haultfœuille 2020; Goodman-Bacon 2021). The estimator is run with year and state by party by office fixed effects, with cohort measured as the first year each state-party-office switched to a more open primary rule treatment (closed to open, closed to open-to-unaffiliated, open to top-four, and open-to-unaffiliated to top-four). The one case where primary type moved in a restrictive direction is removed. The results are in line with those in the main analysis.

Table A.16: **C&S: Effect of Opening Primaries on Voter Turnout**

	Voter Turnout				
	Overall	Black	Asian	Latino	White
	(1)	(2)	(3)	(4)	(5)
Overall ATT	0.078	0.022	0.044	0.032	0.089
	(0.007)	(0.003)	(0.005)	(0.003)	(0.011)
Units	372	372	372	372	372
States	50	50	50	50	50
Years	8	8	8	8	8
Outcome Mean	0.213	0.105	0.080	0.063	0.246

Overall ATT from Callaway-Sant'Anna dynamic aggregation. Weighted average across all treatment cohorts and post-treatment periods using a doubly robust estimator with never-treated units as the control group. Bootstrapped standard errors (1000 iterations) in parentheses. The data range is 2014–2024.

Table A.17: **C&S: Effect of Opening Primaries on Partisan Composition of Electorate**

	Partisan Share			
	Unaffiliated (1)	Third-party (2)	Democratic (3)	Republican (4)
Overall ATT	0.064	0.003	0.003	-0.070
	(0.009)	(0.000)	(0.017)	(0.009)
Units	372	372	372	372
States	50	50	50	50
Years	8	8	8	8
Outcome Mean	0.102	0.008	0.419	0.471

Overall ATT from Callaway-Sant'Anna dynamic aggregation. Weighted average across all treatment cohorts and post-treatment periods using a doubly robust estimator with never-treated units as the control group. Bootstrapped standard errors (1000 iterations) in parentheses. The data range is 2014–2024.

Table A.18: **C&S: Effect of Opening Primaries on Racial Composition of Electorate**

	Racial Share				
	Black (1)	Asian (2)	Latino (3)	Nonwhite (4)	White (5)
Overall ATT	0.012	0.000	0.003	0.009	-0.009
	(0.003)	(0.000)	(0.001)	(0.003)	(0.002)
Units	372	372	372	372	372
States	50	50	50	50	50
Years	8	8	8	8	8
Outcome Mean	0.069	0.018	0.036	0.140	0.860

Overall ATT from Callaway-Sant'Anna dynamic aggregation. Weighted average across all treatment cohorts and post-treatment periods using a doubly robust estimator with never-treated units as the control group. Bootstrapped standard errors (1000 iterations) in parentheses. The data range is 2014–2024.

Table A.19: **C&S: Effect of Opening Primaries on Demographic Composition of Electorate**

	Male Share	Median Age	Demographic Share WC Share	Demographic Share Low-Edu Share	Veteran Share
	(1)	(2)	(3)	(4)	(5)
Overall ATT	0.005	-1.528	0.007	0.006	-0.004
	(0.002)	(0.187)	(0.002)	(0.002)	(0.001)
Units	372	372	367	367	367
States	50	50	50	50	50
Years	8	8	7	7	7
Outcome Mean	0.462	61.678	0.288	0.682	0.056

Overall ATT from Callaway-Sant'Anna dynamic aggregation. Weighted average across all treatment cohorts and post-treatment periods using a doubly robust estimator with never-treated units as the control group. Bootstrapped standard errors (1000 iterations) in parentheses. The data range is 2014–2024.

A.8 Event Studies Estimators

This section studies the plausibility of the parallel trends assumption—that states implementing primary reforms were on similar primary turnout and electorate composition trajectories as states that did not enact reforms. I the show the result of event study estimators for the effect of primary reform on voter turnout (Figure A.3), the share of the electorate that is unaffiliated (Figure A.4), the share of the electorate that is nonwhite (Figure A.5), and the median age of the electorate (Figure A.6). In each case, I filter the sample to offices that hold elections every two years (all congressional primaries, most state legislative primaries, and some state executive primaries) to ensure a consistent treatment timing baseline. I also filter out primary reforms that move in a restrictive direction. Due to the narrow temporal range of the data and the fact that many reforms occurred in the first few years of the panel, I pool effects four or more years prior to the reform and effects four or more years after the reform. Point estimates are relative to election before the primary reform was implemented. I show all reforms in the right panels and just reforms from closed to open to unaffiliated in the left panel.

The event study figures for turnout, unaffiliated share, and age show no evidence of pre-trending and generally show consistently large and statistically significant effects immediately after treatment. There does appear to be some pre-trending in the case of racial demographics: electorates that opened primaries were already on a trajectory of increasing nonwhite primary electorate shares. There is only a statistically significant pre-treatment difference in the closed to open-to-unaffiliated sample and not the full sample of reforms, but taken together this urges for a cautious interpretation of the effect of primary reform on the racial demographics of the electorates.

Figure A.3: **Event Study Estimation of Effect of Opening Primaries on Primary Voter Turnout**. This shows the effect of a switch to a more expansive primary rule on voter turnout in primary elections. The sample is filtered to offices that hold elections every two years and to switches that do not restrict primary participation. Point estimates are relative to the difference in turnout between state-party-offices that change their primary type and those that do not in the election prior to the reform. Effects four or more years prior to the switch are bundled together, as are those at least four years after the switch. The left panel subsets treatment to primary reforms that moved from closed to open to unaffiliated. The right panel includes all primary reforms in an expansive direction. Error bars reflect 95% confidence intervals.

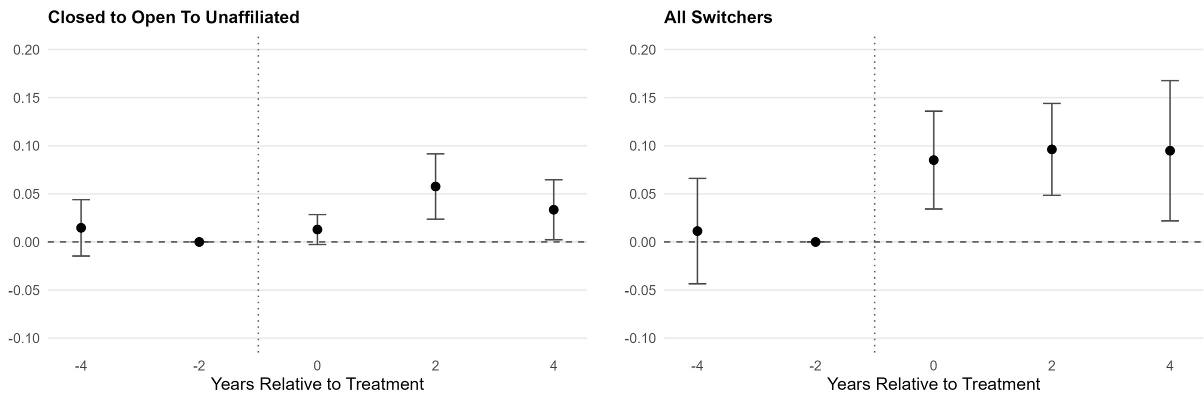


Figure A.4: **Event Study Estimation of Effect of Opening Primaries on Share of Electorate That is Unaffiliated**. This shows the effect of a switch to a more expansive primary rule on the share of the primary electorate that is not affiliated with a political party. The sample is filtered to offices that hold elections every two years and to switches that do not restrict primary participation. Point estimates are relative to the difference in unaffiliated share between state-party-offices that change their primary type and those that do not in the election prior to the reform. Effects four or more years prior to the switch are bundled together, as are those at least four years after the switch. The left panel subsets treatment to primary reforms that moved from closed to open to unaffiliated. The right panel includes all primary reforms in an expansive direction. Error bars reflect 95% confidence intervals.

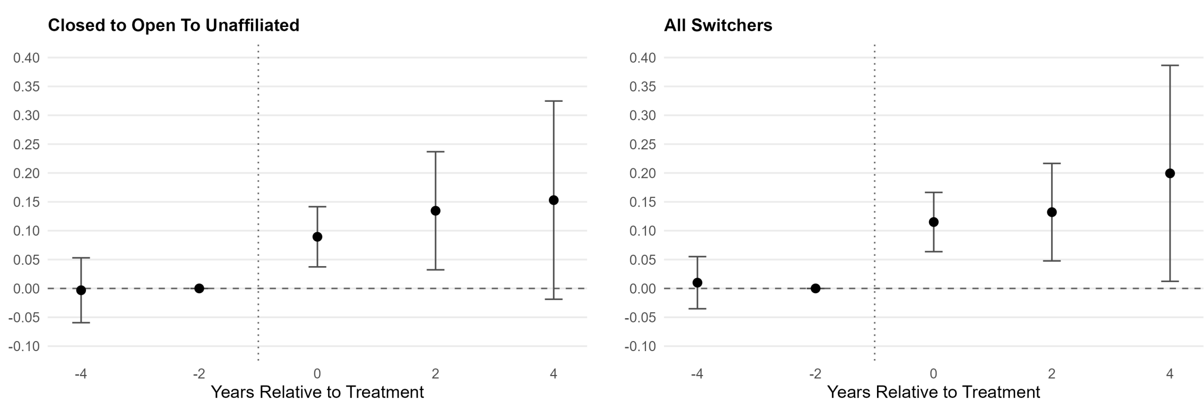


Figure A.5: **Event Study Estimation of Effect of Opening Primaries on Share of Electorate That is Nonwhite**. This shows the effect of a switch to a more expansive primary rule on the share of the primary electorate that is not non-Hispanic white. The sample is filtered to offices that hold elections every two years and to switches that do not restrict primary participation. Point estimates are relative to the difference in nonwhite share between state-party-offices that change their primary type and those that do not in the election prior to the reform. Effects four or more years prior to the switch are bundled together, as are those at least four years after the switch. The left panel subsets treatment to primary reforms that moved from closed to open to unaffiliated. The right panel includes all primary reforms in an expansive direction. Error bars reflect 95% confidence intervals.

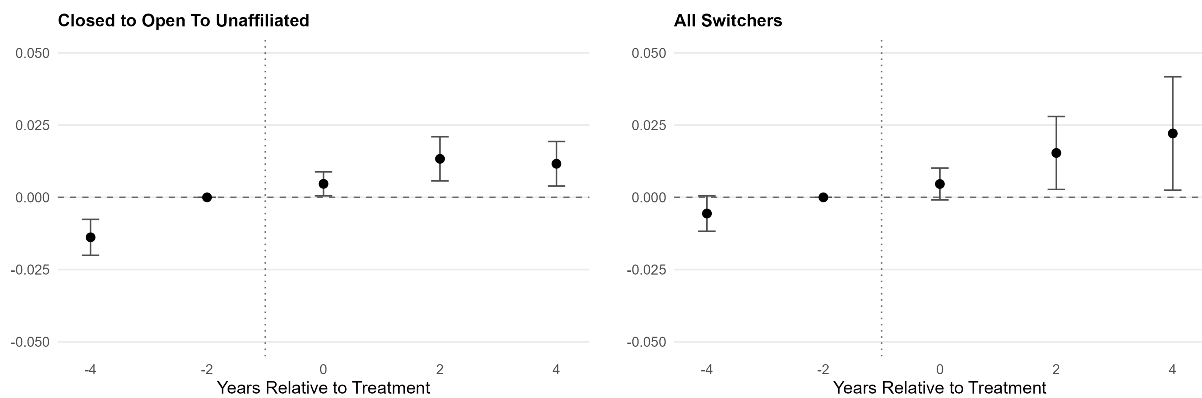
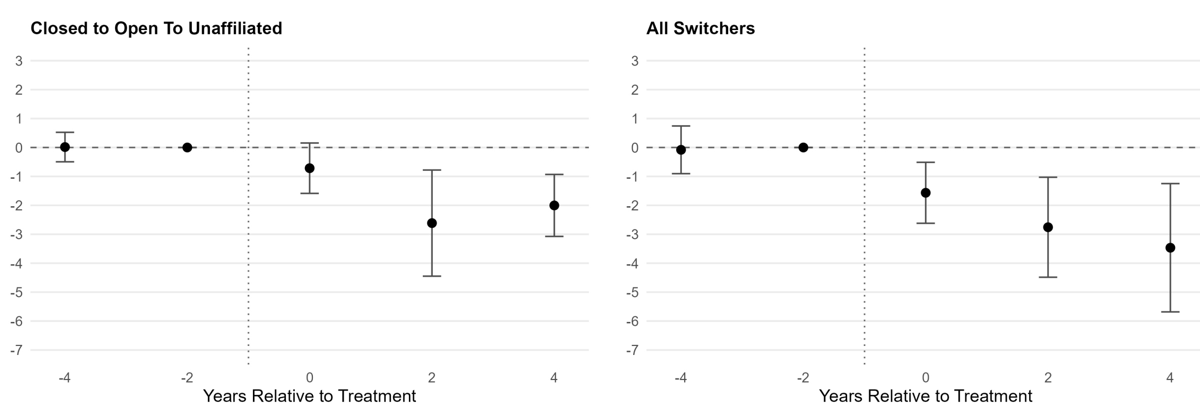


Figure A.6: **Event Study Estimation of Effect of Opening Primaries on Median Age of Primary Electorate**. This shows the effect of a switch to a more expansive primary rule on the median age of primary voters. The sample is filtered to offices that hold elections every two years and to switches that do not restrict primary participation. Point estimates are relative to the difference in median electorate age between state-party-offices that change their primary type and those that do not in the election prior to the reform. Effects four or more years prior to the switch are bundled together, as are those at least four years after the switch. The left panel subsets treatment to primary reforms that moved from closed to open to unaffiliated. The right panel includes all primary reforms in an expansive direction. Error bars reflect 95% confidence intervals.



A.9 Party Registration and Ideological Extremism

Figure A.7 shows the descriptive relationship between party registration and ideological extremity using responses from the 2018 CES, subset to verified registered voters and weighted to the population of registered voters. Table A.20 is similar to Table 11 in the main text, but subsets to verified primary voters.

Table A.20: **Party Registration and Ideological Extremism (Primary Voters Only)**

	Ideological Extremism		
	(1)	(2)	(3)
Democrat	-0.035	-0.010	0.203
	(0.017)	(0.017)	(0.034)
Republican	0.103	0.092	0.313
	(0.017)	(0.018)	(0.034)
3rd Party	-0.208	-0.221	-0.026
	(0.103)	(0.101)	(0.105)
Observations	19637	19633	19633
Outcome Mean	1.115	1.115	1.115
Demographic Controls	No	Yes	Yes
State FEs	No	No	Yes

Standard errors in parentheses. Reference category is Unaffiliated. Sample restricted to verified primary voters. Ideological extremism is measured as absolute distance from the ideological center on a 5-point ideology scale, with 0 indicating moderate, 1 indicating Liberal/Conservative, and 2 indicating Very Liberal/Very Conservative. Data from 2018 Cooperative Election Study filtered to verified registered voters. Regressions are run with weights provided by the survey team. Demographic controls are age, gender, education, race, and family income.

Figure A.7: **Party Registration and Ideological Extremism**. This shows descriptive means for the 2018 CES sample of validated registered voters. Ideological extremism is measured as the absolute distance from the ideological center on a 5-point ideology scale, with 0 indicating moderate, 1 indicating Liberal/Conservative, and 2 indicating Very Liberal/Very Conservative. Data from 2018 Cooperative Election Study filtered to verified registered voters. Means are calculated with weights provided by the survey team.